

MATHEMATICS EXTENSION 2

NSBHS-2019

Inequalities

1 Introduction

Definition 1.1 Every real number x has one and only one of the following properties:

1. $x = 0$,
2. $x > 0$,
3. $-x > 0$ or $x < 0$

Definition 1.2 We write $a > b$, which read " a is greater than b ", if and only if $a - b$ is a positive number; and $a < b$, read " a is less than b ", if and only if $a - b$ is a negative number.

Corollary 1.1 Given two numbers a and b , exactly one of the following assertions is satisfied, $a = b$, $a > b$ or $a < b$.

Corollary 1.2 $a > b$ if and only if $b < a$.

Theorem 1.1 $a > 0$ if and only if a is a positive number, and $a < 0$ if and only if a is a negative number.

Proof: $a > 0$ if and only if $a - 0$ is a positive number (1.2). But $a - 0 = a$. Therefore $a > 0$ if and only if a is a positive number. The proof of the second part is analogous.

Theorem 1.2 If $x > 0$ and $y > 0$ then $x + y > 0$.

Theorem 1.3 If $x > 0$ and $y > 0$ then $xy > 0$.

Theorem 1.4 [Transitive Law of Inequality] If $a > b$ and $b > c$ then $a > c$.

Proof: Since $a > b$ and $b > c$, $a - b$ and $b - c$ are both positive numbers (1.2). Thus $(a - b) + (b - c) > 0$ which implies that $a - c > 0$. Therefore $a > c$. Hence the result.

Theorem 1.5 Let a, b and c be any real numbers. Then $a > b$ if and only if $a + c > b + c$.

Proof: $a > b$ if and only if $a - b > 0$ (by 1.2 and 1.1). But $a - b = (a + c) - (b + c)$. Therefore $a > b$ if and only if $(a + c) - (b + c) > 0$, that is, if and only if $a + c > b + c$.

Theorem 1.6 Let $c > 0$. Then $a > b$ if and only if $ac > bc$.

Proof: Left as an exercise.

Theorem 1.7 Let $c < 0$. Then $a > b$ if and only if $ac < bc$.

Proof: Left as an exercise.

Theorem 1.8 If $a > b$ and $c > d$, then $a + c > b + d$.

Proof: $a - b > 0$ and $c - d > 0$ (1.2). Thus $a - b$ and $c - d$ are positive (1.1). Therefore $(a - b) + (c - d) > 0$. But this may be written $(a + c) - (b + d) > 0$, which implies that $a + c > b + d$.

The proofs of the following four theorems are easy and are left as exercises.

Theorem 1.9 If $a \neq 0$, then $a^2 > 0$.

Theorem 1.10 If $a \neq 0$, $\frac{1}{a}$ has the same sign as a .

Theorem 1.11 If a and b have the same sign, and if $a < b$, then $\frac{1}{a} > \frac{1}{b}$.

Theorem 1.12 Let $a < b$. Then if a and b are both positive, $a^2 < b^2$; and if a and b are both negative, $a^2 > b^2$.

1.1 Useful Identities

1. $a^2 - b^2 = (a - b)(a + b)$
2. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
3. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
4. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
5. $abc = (a + b + c)(ab + bc + ca) - (a + b)(b + c)(a + c)$

1.2 Exercise

Prove the following assertions:

1. $a < 0, \quad b < 0 \implies ab > 0.$
2. $a < 0, \quad b > 0 \implies ab < 0.$
3. $a < b, \quad b < c \implies a < c.$
4. $a < b, \quad c > d \implies a + c < b + d.$
5. $a < b \implies -b < -a.$
6. $a > 0 \implies \frac{1}{a} > 0.$
7. $a < 0 \implies \frac{1}{a} < 0.$
8. $a > 0, \quad b > 0 \implies \frac{a}{b} > 0.$
9. $0 < a < b, \quad 0 < c < d \implies ac < bd.$
10. $a > 1 \implies a^2 > a.$
11. $0 < a < 1 \implies a^2 < a.$

2 AM-GM inequality of two numbers

Consider $a > 0$ and $b > 0$,

$$(a - b)^2 > 0 \quad \text{for all } a \text{ and } b. \quad (1)$$

Expanding (1), we obtain:

$$a^2 - 2ab + b^2 > 0. \quad (2)$$

Hence,

$$a^2 + b^2 > 2ab. \quad (3)$$

Similarly,

$$(\sqrt{a} - \sqrt{b})^2 > 0. \quad (4)$$

leads to

$$a + b > 2\sqrt{ab}. \quad (5)$$

Inequation (5) is called the AM-GM inequality, note that the arithmetic mean of a and b is $\frac{a+b}{2}$ and their geometric mean is $\sqrt{ab}$.

2.1 Application of the AM-GM inequality with two numbers

1. Let $a > 0$ and $b > 0$, prove that

$$(a + b)(1 + ab) \geq 4ab$$

2. If $x > 0$, $y > 0$, $a > 0$ and $b > 0$, show that

$$(ab + xy)(ax + by) \geq 4abxy$$

3. Let $a > 0$ and $b > 0$, prove that

$$a + b + 1 \geq 2\sqrt{a + b}$$

4. Let $a > 0$ and $b > 0$, prove that

$$a^2 + b^2 + 1 \geq ab + a + b$$

5. Let a , b and c be real numbers. Prove that

$$a^2 + b^2 + c^2 \geq ab + ac + bc$$

Deduce that if $a + b + c = 1$, then $ab + bc + ca \leq \frac{1}{3}$.

6. Let $a > 0$ and $b > 0$, prove that

$$\frac{a + b}{2} \leq \sqrt{\frac{a^2 + b^2}{2}}$$

7. Show that if a , b , c and d are positive, then

$$a^4 + b^4 + c^4 + d^4 \geq 4abcd$$

8. if $a > 0$, $b > 0$ and $a + b = t$, show that

(a)

$$\frac{1}{a} + \frac{1}{b} \geq \frac{4}{t}$$

(b)

$$\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{8}{t^2}$$

9. Let $x > 0$, $y > 0$ and $z > 0$, prove that

$$(x+y)(y+z)(z+x) \geq 8xyz$$

10. Let $x > 0$, $y > 0$ and $z > 0$ and given that $x + y + z = 1$, prove that

$$(1-x)(1-y)(1-z) \geq 8xyz$$

11. Three positive numbers a, b, c satisfy the conditions that $a \geq b \geq c$ and that $a + b + c \leq 1$. By considering $(a + b + c)^2$ otherwise, prove that

$$a^2 + 3b^2 + 5c^2 \leq 1$$

12. For $x > 0$, $y > 0$ and $z > 0$, show that

$$x + y + z + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq 6$$

13. For $a > 0$, $b > 0$, $c > 0$ and $d > 0$ and given that $\frac{a+b}{2} \geq \sqrt{ab}$, show that

$$\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$$

14. For $a > 0$, $b > 0$, $c > 0$ and $d > 0$ and given that $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$, show that

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4$$

15. If a , b and c are positive, show that

(a) $a + \frac{1}{a} \geq 2$

(b) $(a+b)(\frac{1}{a} + \frac{1}{b}) \geq 4$ and $(a+b+c)(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}) \geq 9$.

(c) Hence show that

$$\frac{9}{a+b+c} \leq \frac{2}{b+c} + \frac{2}{c+a} + \frac{2}{a+b} \leq \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

16. Let $a_1, a_2, \dots, a_n$ be positive real numbers such that $a_1 a_2 \dots a_n = 1$. Prove that

$$(1+a_1)(1+a_2)\dots(1+a_n) \geq 2^n.$$

17. Prove that $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$, where a , b and c are real.

18. Prove that $x^2 + x + 1 > 0$ for all real x . Hence, or otherwise prove that:

(a) $a^2 + ab + b^2 > 0$.

(b) $a^4 + b^4 \geq a^3b + ab^3$.

19. Given that $x^2 + y^2 + z^2 \geq xy + yz + zx$ for $x > 0$, $y > 0$ and $z > 0$. Prove that

$$(x+y+z)^2 \geq 3(xy+yz+zx).$$

20. Given a , b and c are positive real numbers. Prove that

$$a^2b^2 + b^2c^2 + c^2a^2 \geq abc(a+b+c).$$

21. Let $x_0 > x_1 > x_2 > \dots > x_n$ be real numbers. Prove that

$$x_0 + \frac{1}{x_0 - x_1} + \frac{1}{x_1 - x_2} + \dots + \frac{1}{x_{n-1} - x_n} \geq x_n + 2n.$$

22. Prove that if $a + b = 1$, then $a^3 + b^3 \geq \frac{1}{4}$.

23. If a and b are real numbers, show that $a^4 + b^4 \geq ab(a^2 + b^2)$.

3 AM-GM inequality of three numbers

Consider $a > 0$, $b > 0$ and $c > 0$, expand the following product:

$$(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

, we obtain

$$a^3 + b^3 + c^3 - 3abc.$$

Hence

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

Using (5), we obtain: $a^2 + b^2 \geq 2ab$, $a^2 + c^2 \geq 2ac$ and $b^2 + c^2 \geq 2bc$. Hence

$$a^2 + b^2 + c^2 \geq ab + bc + ca.$$

Since $a + b + c \geq 0$ and $a^2 + b^2 + c^2 - ab - bc - ca \geq 0$, we obtain that

$$a^3 + b^3 + c^3 - 3abc \geq 0.$$

Hence

$$a^3 + b^3 + c^3 \geq 3abc.$$

By a suitable substitution, we obtain

$$a + b + c \geq 3\sqrt[3]{abc}.$$

3.1 Application of the AM-GM inequality with three numbers

1. Given a , b and c are positive real numbers. Prove that $\frac{1}{3}(a^3 - b^3)(a - b) \geq ab(a - b)^2$.
2. Given a , b and c are positive real numbers such that $abc = 1$. Prove that $a^2 + b^2 + c^2 \geq a + b + c$.
3. Given a , b and c are positive real numbers. Prove that

$$\frac{a^3}{b} + \frac{b^3}{c} + \frac{c^3}{a} \geq ab + bc + ca.$$

4. Given a , b and c are positive real numbers. Prove that

$$(a^2b + b^2c + c^2a)(ab^2 + bc^2 + ca^2) \geq 9a^2b^2c^2.$$

5. (Nesbitt's Inequality) Given a , b and c are positive real numbers. Prove that

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}.$$

6. Given a , b and c are positive real numbers such that $abc = 1$. Prove that

$$\frac{1+ab}{1+a} + \frac{1+bc}{1+b} + \frac{1+ac}{1+c} \geq 3.$$

7. (Ireland 1998) Given a , b and c are positive real numbers. Prove that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 2\left(\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a}\right) \geq \frac{9}{a+b+c}.$$

8. Given a , b and c are positive real numbers. Prove that

$$\frac{1}{b(a+b)} + \frac{1}{c(b+c)} + \frac{1}{a(c+a)} \geq \frac{27}{2(a+b+c)^2}.$$

9. Given a, b and c are positive real numbers such that $abc = 1$. Prove that

$$\frac{a}{(a+1)(b+1)} + \frac{b}{(b+1)(c+1)} + \frac{c}{(c+1)(a+1)} \geq \frac{3}{4}.$$

10. Given a, b and c are positive real numbers such that $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = 1$. Prove that

$$abc \geq 8.$$

11. $\triangle ABC$ has side length a, b and c . If $a^2 + b^2 + c^2 = ab + bc + ca$, show that $\triangle ABC$ is an equilateral triangle.

4 Mixed Problems

1. If m, n, p and q are positive numbers. Show that:

(a) $m + n \geq 2\sqrt{mn}$

(b) $(m+n)(n+p)(m+p) \geq 8mnp$

(c) $\frac{m}{n} + \frac{n}{p} + \frac{p}{q} + \frac{q}{m} \geq 4.$

2. Let $a > 0$ and $b > 0$. Show that $a^3 + b^3 \geq a^2b + ab^2$.

3. Given that a, b and c are real positive numbers.

(a) Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$.

(b) Show that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a+b+c}$.

(c) Given that $a^2 + b^2 + c^2 = 9$, prove that

$$\frac{1}{1+ab} + \frac{1}{1+bc} + \frac{1}{1+ac} \geq \frac{3}{4}.$$

4. Given that a, b and c are real numbers.

(a) Show that $a^2 + 9b^2 \geq 6ab$.

(b) Hence or otherwise, show that $a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ca)$

(c) Hence if $a > b > c > 0$, show that $a^2 + 5b^2 + 9c^2 > 9bc$.

5. (NSBHS 2019, Task 2)

Given that a, b and c are real positive numbers such that $a + b + c = 1$ and $a + b + c \geq 3^3\sqrt{abc}$

(a) If x, y and z are all positives. Show that $\frac{1}{xy} + x + y \geq 3$.

(b) Hence or otherwise, prove that

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} \geq 4.$$

6. Given that a and b are real numbers.

- (a) Prove that $a^2 + b^2 \geq 2ab$.
- (b) If $x > 0$ and $y > 0$, prove that $\frac{x}{y} + \frac{y}{x} \geq 2$.
- (c) Prove by induction, or otherwise that

$$(x_1 + \cdots + x_n)\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right) \geq n^2,$$

where x_i are all real and positives.

7. Given a, b and c are positive real numbers. Prove that

$$(a^3 + 2)(b^3 + 2)(c^3 + 2) \geq (a + b + c)^3.$$

$$\frac{a}{(a+1)(b+1)} + \frac{b}{(b+1)(c+1)} + \frac{c}{(c+1)(a+1)} \geq \frac{3}{4}.$$

8. (Jomaa 2019) Given that a, b and c are positive real numbers such that $a + b + c = 1$, prove that

$$\frac{2}{(1+a)^2} + \frac{2}{(1+b)^2} + \frac{2}{(1+c)^2} \leq \frac{1}{(a+b)(a+c)(b+c)}.$$

9. Given that a_1, b_1, a_2 , and b_2 are real numbers.

- (a) Prove that $(a_1^2 + a_2^2)(b_1^2 + b_2^2) \geq (a_1b_1 + a_2b_2)^2$
- (b) Hence prove by induction that

$$(a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2) \geq (a_1b_1 + \cdots + a_nb_n)^2,$$

where $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are real numbers.

- (c) Hence or otherwise, prove that

$$\frac{a^2}{y+z} + \frac{b^2}{x+z} + \frac{c^2}{x+y} \geq \frac{a+b+c}{2},$$

where a, b, c, x, y and z are positive real numbers and $a + b + c = x + y + z$.

10. (NSBHS, Mar 2019) Given a, b and c are positive real numbers.

- (a) Prove that $a^2 + b^2 \geq 2ab$.
- (b) Given that

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca),$$

prove that $\frac{a^3 + b^3 + c^3}{3} \geq abc$.

- (c) Explain why $\frac{a^3 + b^3 + 1}{3} \geq ab$.
- (d) Hence or otherwise, prove that

$$a^3 + b^3 + c^3 + 1 \geq ab + bc + ca + abc.$$

11. Given a, b and c are positive real numbers.

(a) Prove that $a^2 + b^2 \geq 2ab$.

(b) Show that

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca).$$

(c) Show that $a^2 + b^2 + c^2 \geq ab + bc + ca$.

(d) Deduce that $a^3 + b^3 + c^3 \geq 3abc$.

(e) Show that

$$2(a^3 + b^3 + c^3) + \frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} \geq 3abc + 6.$$

(f) Show that

$$a + b + c + a^2 + b^2 + c^2 + a^3 + b^3 + c^3 \geq 3(ab + bc + ca).$$

12. Given a, b and c are positive real numbers such that $abc = 1$. Prove that

$$a^3 + b^3 + c^3 + (ab)^3 + (bc)^3 + (ca)^3 \geq 2(a^2b + b^2c + c^2a).$$

13. Given a, b and c are positive real numbers. Prove that

$$\frac{a^2}{b^2} + \frac{b^2}{c^2} + \frac{c^2}{a^2} \geq \frac{b}{a} + \frac{c}{b} + \frac{a}{c}.$$

14. Given a, b and c are positive real numbers. Prove that

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{a + b + c}{abc}.$$

15. Given that a, b, c, x, y and z are real numbers.

(a) Prove that $(a^2 + 1)(b^2 + 1) \geq (a + b)^2$.

(b) Prove that $(a^2 + 1)(b^2 + 1) \geq (ab + 1)^2$.

(c) Prove that $(a^2 + b^2)(c^2 + d^2) \geq (ac + bd)^2$.

(d) Prove that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) \geq (ax + by + cz)^2$.

5 HSC inequalities questions

1. (2015 HSC, 15c)

For positive real numbers x and y , $\sqrt{xy} \leq \frac{x + y}{2}$.

(a) Prove $\sqrt{xy} \leq \sqrt{\frac{x^2 + y^2}{2}}$, for positive real numbers x and y .

(b) Prove $\sqrt[4]{abcd} \leq \sqrt{\frac{a^2 + b^2 + c^2 + d^2}{4}}$, for positive real numbers a, b, c and d .

2. (2014 HSC, 15a)

Three positive real numbers a, b and c are such that $a + b + c = 1$ and $a \leq b \leq c$. By Considering the expansion of $(a + b + c)^2$, or otherwise, show that

$$5a^2 + 3b^2 + c^2 \leq 1.$$

3. (2012 HSC, 15a)

- (a) Prove that $\sqrt{ab} \leq \frac{a+b}{2}$, where $a \geq 0$ and $b \geq 0$.
- (b) If $1 \leq x \leq y$, show that $x(y-x+1) \geq y$.
- (c) Let n and j be positive integers with $1 \leq j \leq n$. Prove that

$$\sqrt{n} \leq \sqrt{j(n-j+1)} \leq \frac{n+1}{2}.$$

- (d) For integers $n \geq 1$, prove that

$$(\sqrt{n})^n \leq n! \leq \left(\frac{n+1}{2}\right)^n.$$

4. (2011 HSC, 5b)

If p, q and r are positive real numbers and $p+q \geq r$, prove that

$$\frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \geq 0.$$

5. (2004 HSC, 7a)

- (a) Let a be a positive real number. Show that $a + \frac{1}{a} \geq 2$.
- (b) Let n be a positive integer and $a_1, a_2, \dots, a_n$ be n positive real numbers. Prove by induction that

$$(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \geq n^2.$$

- (c) Hence show that $\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta \geq 9 \cos^2 \theta$.

6. (2003 HSC, 6c)

- (a) Let x and y be real numbers such that $x \geq 0$ and $y \geq 0$. Prove that $\frac{x+y}{2} \geq \sqrt{xy}$.
- (b) Suppose that a, b, c are real numbers. Prove that $a^4 + b^4 + c^4 \geq a^2b^2 + a^2c^2 + b^2c^2$.
- (c) Show that $a^2b^2 + a^2c^2 + b^2c^2 \geq a^2bc + b^2ac + c^2ab$.
- (d) Deduce that if $a+b+c=d$, then $a^4 + b^4 + c^4 \geq abcd$.

7. (2001 HSC, 8a)

- (a) Show that $2ab \leq a^2 + b^2$ for all real numbers a and b .
Hence deduce that $3(ab+bc+ca) \leq (a+b+c)^2$ for all real numbers a, b and c .
- (b) Suppose that a, b and c are the sides of a triangle. Explain why $(b-c)^2 \leq a^2$.
Deduce that $(a+b+c)^2 \leq 4(ab+bc+ca)$.

6 Trial exams inequality problems

1. (NSBHS 2019)

Given that a, b and c are positive real numbers.

- (a) Prove that $a^2 + (bc)^2 \geq 2abc$.
- (b) Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$.
- (c) Prove that $a^2(1 + b^2) + b^2(1 + c^2) + c^2(1 + a^2) \geq 6abc$.
- (d) Hence, or otherwise prove that $a^2(1 + a^2) + b^2(1 + b^2) + c^2(1 + c^2) \geq 6abc$.

2. (NSGHS 2018, 15a)

- (a) Prove that, if $c \geq a$ and $d \geq b$ then $ab + cd \geq bc + ad$.
- (b) Use part (2a) to show that if $x \geq y$, then $x^2 + y^2 \geq 2xy$.
- (c) Without the use of further algebra, explain why the result in (2b) is in fact true for all the values of x and y .

3. (CSSA 2018, 13d)

- (a) If a, b and c are positive real numbers, show that

$$a^2 \geq (a + b - c)(a + c - b).$$

- (b) Hence show that

$$abc \geq (a + b - c)(b + c - a)(a + c - b).$$

4. (BBHS 2018, 16b)

If $m > 0, n > 0, p > 0, q > 0$, show that

- (a) $m+n \geq 2\sqrt{mn}$.
- (b) $(m+n)(n+p)(p+m) \geq 8mnp$.
- (c) $\frac{m}{n} + \frac{n}{p} + \frac{p}{q} + \frac{q}{m} \geq 4$.

5. (Caringbah HS 2018, 15c)

If $a > 0$ and $b > 0$, show that $a^3 + b^3 \geq a^2b + ab^2$.

6. (Hornsby Girls 2018, 16b)

It is known that n is a positive integer, where $n \geq 2$ and that a and b are real and positive.

- (a) Prove that $\frac{a+b}{2} \geq \sqrt{ab}$.

It is known that for positive real numbers $a_1, a_2, \dots, a_n$ that $\sqrt[n]{a_1 a_2 a_3 \dots a_n} \leq \frac{a_1 + a_2 + \dots + a_n}{n}$.

- (b) Prove that $n! \leq \left(\frac{n+1}{2}\right)^n$.

7. (JRAHS 2018, 12c)

(a) Prove that for all real a and b ,

$$a^2 + b^2 \geq 2ab.$$

(b) Prove that, for any positive, real x and y ,

$$\frac{x}{y} + \frac{y}{x} \geq 2.$$

(c) Prove by induction, or otherwise, that

$$(x_1 + x_2 + \cdots + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n} \right) \geq n^2,$$

where the x_i are all real and positive.

8. (NSBHS 2018, 16b)

Given that a, b and c are real positive numbers such that

$$a + b + c = 1 \quad \text{and} \quad a + b + c \geq 3^3 \sqrt{abc}$$

(a) If x, y and z are all positives. Show that $\frac{1}{xy} + x + y \geq 3$.

(b) Hence or otherwise, prove that

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} \geq 4.$$

9. (Ind 2017,12c)

If $a > 0, b > 0, c > 0$ and $d > 0$ are real numbers, show that

$$(a) \quad a^2 + b^2 + c^2 + d^2 \geq \frac{2}{3}(ab + ac + ad + bc + bd + cd)$$

$$(b) \quad \left(\frac{a + b + c + d}{4} \right)^2 \geq \frac{1}{6}(ab + ac + ad + bc + bd + cd)$$

10. (JRAHS 2017, 15c)

Given that

$$\frac{1}{3}(x + y + z) \geq (xyz)^{\frac{1}{3}},$$

for all x, y, z non-negative real numbers.

(a) Show that

$$xy + yz + zx \geq 3(xyz)^{\frac{2}{3}}.$$

(b) Hence show that

$$xyz \leq (a - 1)^3.$$

If $1 + (x + y + z) + (xy + yz + zx) + xyz = a^3$, where $a \geq 2$ is a real number.

11. (Hornsby GHS 2017, 16b)

(a) Prove that if a and b are any two positive real numbers, then

$$ab \leq \left(\frac{a+b}{2}\right)^2 \leq \frac{a^2+b^2}{2}.$$

(b) Given also that $a+b=1$, prove that

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 \geq \frac{25}{2}.$$

12. (Ascham 2016, 12b)

Use the fact that $x^2 + y^2 \geq 2xy$, or otherwise, to prove that

(a) $\frac{a}{b} + \frac{b}{a} \geq 2$ [You may assume $a, b \geq 0$.]

(b) $p^2 + q^2 + r^2 \geq pq + qr + rp$

(c) $p^3 + q^3 \geq pq(p+q)$ (assume that $p, q \geq 0$)
and hence

(d) $2(p^3 + q^3 + r^3) \geq pq(p+q) + qr(q+r) + rp(r+p)$.

13. (Cringbah 2016, 16c)

If a and b are positive numbers, prove that $a+b \geq 2\sqrt{ab}$.

Hence, or otherwise, prove that when a, b and c are all positive numbers and $a+b+c=1$, then

$$\left(\frac{1}{a}-1\right)\left(\frac{1}{b}-1\right)\left(\frac{1}{c}-1\right) \geq 8.$$

14. (Hunters Hill 2016, 14b)

(a) Prove that $a^2 + b^2 \geq 2ab$.

(b) Hence, or otherwise, Prove that

$$(p+2)(q+2)(p+q) \geq 16pq.$$

where p and q are positive real numbers.

15. (Ind 2016, 13b)

If $a > 0$, $b > 0$ and $c > 0$ are real numbers, show that

(a)

$$(a+b+c)^2 \geq 3(ab+bc+ca).$$

(b)

$$(ab+bc+ca)^2 \geq 3abc(a+b+c).$$

(c)

$$(a+b+c)^3 \geq 27abc.$$

16. (Knox Grammar 2016, 13d)

- (a) using the identity $(p+q)^2 = (p-q)^2 + 4pq$, show that for $p, q > 0$

$$\frac{p+q}{2} \geq \sqrt{pq}.$$

- (b) Hence show that if p, q, r and s are greater than zero then

$$\frac{p+q+r+s}{4} \geq \sqrt[4]{pqrs}.$$

17. (NSBHS 2016, 16a)

- (a) Given that $\frac{a+b}{2} \geq \sqrt{ab}$.

Prove that $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$.

- (b) Using part (17a) and the fact that $\frac{a+b+c}{3} = \frac{1}{4} \left(a+b+c + \frac{a+b+c}{3} \right)$,

Prove that $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$

18. (NSGHS 2016, 14c)

- (a) If a and b are real numbers, prove that $a^2 + b^2 \geq 2ab$.

- (b) Show that $a^2 + b^2 + c^2 \geq ab + bc + ca$, where a, b and c are real.

- (c) Hence or otherwise show that the equation $x^3 - 3x^2 + 4x + k = 0$ can not have three real roots for any real constant k .

19. (Penrith HS 2016, 14d)

If $0 < x < y < \frac{1}{2}$. Prove that

$$\sqrt{xy} < x + y < \sqrt{x+y}.$$

20. (Sam el Hosri 2016, 15a)

- (a) Given that a, b and c are three positive integers, show that

$$a^2 + b^2 + c^2 \geq ab + bc + ca.$$

- (b) Given that

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac).$$

Show that

$$\frac{p^2}{q^2} + \frac{q^2}{r^2} + \frac{r^2}{p^2} \geq \frac{p}{r} + \frac{q}{p} + \frac{r}{q} \geq 3,$$

where p, q and r are positive integers.

21. (Western Region 2016, 14d)

- (a) If x, y, z are real and unequal, show that $x^2 + y^2 \geq 2xy$

and hence deduce that

$$x^2 + y^2 + z^2 \geq xy + yz + zx.$$

- (b) if $x + y + z = 3$, show that $xy + yz + zx < 3$.

22. (ACE 2015, 16b)

Show that

$$(a+b+c)^2 \leq 3(a^2 + b^2 + c^2).$$

23. (Ind 2015, 16c)

a, b and c are real numbers such that $a > b > c > 1$.

(a) Show that $a^a b^b c^c > a^b b^c c^a$.

(b) Hence show that $a \ln a + b \ln b + c \ln c > b \ln a + c \ln b + a \ln c$.

24. (Knox Grammar 2015, 15c)

(a) It can be shown that for positive real numbers a and b that

$$a^2 + b^2 \geq 2ab \quad (\text{DO NOT PROVE THIS}).$$

Hence show for positive real numbers a, b, c and d that

$$3(a^2 + b^2 + c^2 + d^2) \geq 2(ab + ac + ad + bc + bd + cd).$$

(b) Hence show for positive real numbers a, b, c and d that if $a + b + c + d = 1$ then

$$ab + ac + ad + bc + bd + cd \leq \frac{3}{8}.$$

25. (NSGHS 2015, 16b)

(a) Show that if a and b are real, then $a^2 + b^2 \geq 2ab$.

(b) Hence show that if a, b, c and d are real, then $a^4 + b^4 + c^4 + d^4 \geq 4abcd$.

26. (Sam el Hosri 2015, 16c)

The three positive real numbers p, q , and r are such that

$$(p+2)(q+2)(r+2) = 64.$$

Given that $a + b + c \geq 3\sqrt{abc}$, for any three real positive numbers a, b and c , show that

$$pqr \leq 8.$$

27. (Ind 2014, 15b)

(a) If $a > 0$ is a real number, show that $a + \frac{1}{a} \geq 2$.

(b) Hence show that if $a > 0, b > 0, c > 0$ are real numbers, then

i.

$$\frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c} \geq 6.$$

ii.

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

28. (SBHS 2014, 14c)

In each of the following parts, $x, y, z, w, a, b, c, d > 0$:

(a) Show that $(x+y)^2 \geq 4xy$.

(b) Show that $[(x+y)(z+w)]^2 \geq 16xyzw$.

(c) Deduce that $\frac{x+y+z+w}{4} \geq \sqrt[4]{xyzw}$.

(d) Hence show that (using (28c)):

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4.$$

29. (ACE 2013, 16c)

Show that $\frac{a^2 + b^2}{a - b} \geq 2\sqrt{2}$ for $a > b$ and $ab > 1$.

30. (Ind 2013, 16b)

(a) If a and b are positive real numbers, show that $a^2 - ab + b^2 \geq \left(\frac{a+b}{2}\right)^2$.

(b) In $\triangle ABC$, if $\angle BCA \geq 60^\circ$ show that $c^2 \geq a^2 - ab + b^2$ and hence deduce that $2c^3 \geq a^3 + b^3$ with equality if and only if $\triangle ABC$ is equilateral.

31. (NHBHS 2013, 16b)

(a) By squaring, or otherwise, show that for $k \geq 0$,

$$2k + 3 \geq 2\sqrt{k+2}\sqrt{k+1}.$$

(b) By decomposing $2k + 3$ and factorising $2\sqrt{k+2}\sqrt{k+1} - 2(k+1)$ show that for $k \geq 1$

$$\frac{1}{\sqrt{k+1}} > 2(\sqrt{k+2} - \sqrt{k+1}).$$

(c) Hence, or otherwise, show for $n \geq 1$,

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} > 2(\sqrt{n+1} - 1).$$

32. (NSGHS 2013, 13a)

(a) Show that $x^2 + y^2 \geq xy$, where x and y are real numbers.

(b) If $x + y = 3z$, show that $x^2 + y^2 \geq 3z^2$.

33. (Penrith 2013, 15d)

(a) Prove that $x^2 + x + 1 \geq 0$ for all real x .

(b) Hence or otherwise, prove that $a^2 + ab + b^2 \geq 0$.

34. (Sam el Hosri 2013, 15a)

(a) Show that $a + \frac{1}{a} \geq 2$ for $a > 0$.

(b) Given a, b, c are three positive real numbers such that $a + b + c = 1$. Show that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 5.$$

35. (NSGHS 2012, 15c)

(a) Show that $a^2 + b^2 \geq 2ab$ for any values of a and b .

(b) Hence show that $\tan^2 \theta + \cos^2 \theta + \operatorname{cosec}^2 \theta \geq \sin \theta + \sec \theta + \cot \theta$ for all values of θ .

36. (SBHS 2012, 15b)

For positive real numbers x and y

(a) Prove that $\frac{x+y}{2} \geq \sqrt{xy}$.

When is there equality?

(b) Hence by considering $\frac{1}{a} + \frac{1}{b}$, or otherwise, prove that $\frac{2ab}{a+b} \leq \sqrt{ab}$ for positive real numbers a, b .

(c) Hence, or otherwise prove that $\frac{1}{x-1} + \frac{1}{x} + \frac{1}{x+1} \geq \frac{3}{x}$ for any $x > 1$.

(d) If $H = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}$, where n is an integer $n > 1$, use (36c) to show that H has no limit as $n \rightarrow \infty$.

37. (Manly SC 2011, 6b)

(a) Show that $a^2 + b^2 \geq 2ab$ where a and b are distinct positive real numbers.

(b) Hence show that $a^2 + b^2 + c^2 \geq ab + ac + bc$.

(c) Hence show that $\sin^2 \alpha + \cos^2 \alpha \geq \sin 2\alpha$.

(d) Hence show that $\sin^2 \alpha + \cos^2 \alpha + \tan^2 \alpha \geq \sin \alpha - \cos \alpha + \sec \alpha + \frac{1}{2} \sin 2\alpha$.

38. (NSGHS 2011, 8b)

(a) If $a > 0, b > 0$ and $c > 0$, show that $a^2 + b^2 \geq 2ab$ and hence deduce that $a^2 + b^2 + c^2 \geq ab + bc + ca$.

(b) If $a + b + c = 9$, show that $ab + bc + ca \leq 27$ and $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{27}{abc}$.

39. (CSSA 2009, 6b)

(a) Given that $x > 0$ and $y > 0$ show that

$$x + y \geq 2\sqrt{xy}.$$

(b) Hence show that for $x > 0, y > 0, z > 0$ and $w > 0$

$$x + y + z + w \geq 4\sqrt[4]{xyzw}.$$

(c) Consider x, y, z and $w = \frac{x+y+z}{3}$. Apply the result in (39b) to show that

$$\frac{x+y+z}{3} \geq \sqrt[3]{xyz}.$$

Application of the AM-GM inequality with two numbers.

1. $(a+b)(1+ab) \geq 4ab$

$$(\sqrt{a}-\sqrt{b})^2 \geq 0 \therefore a+b \geq 2\sqrt{ab} \quad (1)$$

$$(\sqrt{ab}-1)^2 \geq 0 \therefore 1+ab \geq 2\sqrt{ab} \quad (2)$$

$$(1) \times (2) \therefore (a+b)(1+ab) \geq 4ab.$$

2. $(ab+xy)(ax+by) \geq 4abxy$

$$(\sqrt{ab}-\sqrt{xy})^2 \geq 0 \therefore ab+xy \geq 2\sqrt{abxy} \quad (1)$$

$$(\sqrt{ax}-\sqrt{by})^2 \geq 0 \therefore ax+by \geq 2\sqrt{abxy} \quad (2)$$

$$(1) \times (2) \therefore (ab+xy)(ax+by) \geq 4abxy$$

3. $a+b+1 \geq 2\sqrt{a+b}$

$$(\sqrt{a+b}-1)^2 \geq 0 \therefore a+b+1 \geq 2\sqrt{a+b}.$$

4. $a^2+b^2+1 \geq ab+a+b$

$$(a-b)^2 + (a-1)^2 + (b-1)^2 \geq 0$$

$$a^2-2ab+b^2+a^2-2a+1+b^2-2b+1 \geq 0$$

$$2(a^2+b^2+1) \geq 2(ab+a+b)$$

$$\therefore a^2+b^2+1 \geq ab+a+b.$$

5. $a^2+b^2+c^2 \geq ab+ac+bc$

$$(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$$

$$a^2-2ab+b^2+b^2-2bc+c^2+c^2-2ac+a^2 \geq 0$$

$$2(a^2+b^2+c^2) \geq 2(ab+bc+ca)$$

$$\therefore a^2+b^2+c^2 \geq ab+bc+ca.$$

$$a+b+c=1 \therefore (a+b+c)^2=1 \therefore$$

$$a^2+b^2+c^2+2(ab+bc+ca)=1 \geq ab+bc+ca+2(ab+bc+ca)$$

$$\therefore 3(ab+bc+ca) \leq 1 \therefore ab+bc+ca \leq \frac{1}{3}.$$

$$\begin{aligned}
 6. \quad & (a-b)^2 \geq 0 \\
 & (a-b)^2 + (a+b)^2 \geq (a+b)^2 \\
 & 2(a^2+b^2) \geq (a+b)^2 \\
 & \frac{a^2+b^2}{2} \geq \left(\frac{a+b}{2}\right)^2 \\
 & \sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad & (a^2-b^2)^2 \geq 0 \therefore a^4+b^4 \geq 2a^2b^2 \quad (1) \\
 & \text{Similarly } c^4+d^4 \geq 2c^2d^2 \quad (2)
 \end{aligned}$$

$$(1) + (2) \therefore a^4+b^4+c^4+d^4 \geq 2(a^2b^2+c^2d^2)$$

$$\text{But } (ab-cd)^2 \geq 0 \therefore a^2b^2+c^2d^2 \geq 2abcd$$

$$\text{Now } a^4+b^4+c^4+d^4 \geq 2(2abcd) = 4abcd.$$

$$8. \quad (a) \quad (a-b)^2 \geq 0 \therefore (a+b)^2 - 4ab \geq 0$$

$$\frac{(a+b)^2}{ab(a+b)} - \frac{4ab}{ab(a+b)} \geq 0$$

$$\frac{a+b}{ab} - \frac{4}{t} \geq 0$$

$$\frac{1}{a} + \frac{1}{b} \geq \frac{4}{t}$$

$$\left. \begin{aligned}
 & \text{Alternatively} \\
 & (a+b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4 \\
 & \therefore \frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}
 \end{aligned} \right\}$$

(b)

$$(a+b)(a+b)\left(\frac{1}{a^2} + \frac{1}{b^2}\right) \geq 4ab \times \frac{2}{ab} = 8$$

$$\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{8}{t^2}$$

$$9. \quad \left. \begin{aligned}
 x+y & \geq 2\sqrt{xy} \\
 y+z & \geq 2\sqrt{yz} \\
 z+x & \geq 2\sqrt{zx}
 \end{aligned} \right\} (x+y)(y+z)(z+x) \geq 8xyz.$$

10. $x+y+z=1$

$$1-x = y+z$$

$$1-y = x+z$$

$$1-z = x+y$$

$$1-x = y+z \geq 2\sqrt{yz}$$

$$1-y = x+z \geq 2\sqrt{xz}$$

$$1-z = x+y \geq 2\sqrt{xy}$$

$$(1-x)(1-y)(1-z) \geq 8xyz$$

11.

$$a \geq b \geq c$$

$$a \geq b \therefore ab \geq b^2$$

$$a \geq c \therefore ac \geq c^2$$

$$b \geq c \therefore bc \geq c^2$$

$$a+b+c \leq 1 \therefore (a+b+c)^2 \leq 1$$

$$\therefore a^2 + b^2 + c^2 + 2(ab+bc+ca) \leq 1 \quad (1)$$

$$2(ab+bc+ca) \geq 2b^2 + 2c^2 + 2c^2 \quad (2)$$

Combining (1) + (2) we obtain:

$$\therefore a^2 + 3b^2 + 5c^2 \leq 1$$

12.

$$x+y+z + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq 6$$

$$x + \frac{1}{x} \geq 2$$

$$y + \frac{1}{y} \geq 2$$

$$z + \frac{1}{z} \geq 2$$

$$x+y+z + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq 6.$$

13.

$$\frac{a+b}{2} \geq \sqrt{ab}, \text{ similarly } \frac{c+d}{2} \geq \sqrt{cd}$$

$$\frac{a+b+c+d}{4} = \frac{1}{2} \left(\frac{a+b}{2} + \frac{c+d}{2} \right) \geq \frac{1}{2} (\sqrt{ab} + \sqrt{cd}) \geq \sqrt{\sqrt{ab}\sqrt{cd}} = \sqrt[4]{abcd}$$

14. Given that $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$

$$\frac{\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}}{4} \geq \sqrt[4]{\frac{a}{b} \frac{b}{c} \frac{c}{d} \frac{d}{a}} = 1$$

$$\therefore \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4.$$

15. (a) $(\sqrt{a} - \frac{1}{\sqrt{a}})^2 \geq 0 \therefore a - 2 + \frac{1}{a} \geq 0 \therefore a + \frac{1}{a} \geq 2$

(b) $(a+b)(\frac{1}{a} + \frac{1}{b}) \geq 4$?

$$a+b \geq 2\sqrt{ab}$$

$$\frac{1}{a} + \frac{1}{b} \geq 2\sqrt{\frac{1}{a} \frac{1}{b}} = \frac{2}{\sqrt{ab}}$$

$$(a+b)(\frac{1}{a} + \frac{1}{b}) \geq 2\sqrt{ab} \times \frac{2}{\sqrt{ab}} = 4.$$

$(a+b+c)(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}) \geq 9$?

$$(a+b+c)(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$$

$$= (a+b)(\frac{1}{a} + \frac{1}{b}) + \frac{a}{c} + \frac{b}{c} + \frac{c}{a} + \frac{c}{b} + \frac{c}{c}$$

$$= (a+b)(\frac{1}{a} + \frac{1}{b}) + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b} + 1$$

$$\geq 4 + 2 + 2 + 1 = 9.$$

(c) $2(a+b+c)\left[\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}\right]$

$$= (b+c+c+a+a+b)\left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}\right) \geq 9$$

By the previous exercise. (see 15).

$$\therefore \frac{2}{b+c} + \frac{2}{c+a} + \frac{2}{a+b} \geq \frac{9}{a+b+c}.$$

$$(a+b)(\frac{1}{a} + \frac{1}{b}) \geq 4, (a+c)(\frac{1}{a} + \frac{1}{c}) \geq 4, (b+c)(\frac{1}{b} + \frac{1}{c}) \geq 4$$

$$\therefore \frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}, \frac{1}{a} + \frac{1}{c} \geq \frac{4}{a+c}, \frac{1}{b} + \frac{1}{c} \geq \frac{4}{b+c}$$

$$2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq \frac{4}{a+b} + \frac{4}{a+c} + \frac{4}{b+c} \therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{2}{a+b} + \frac{2}{b+c} + \frac{2}{c+a}.$$

$$16. \quad 1 + a_i \geq 2\sqrt{a_i}, \quad i=1, \dots, n$$

$$(1 - \sqrt{a_i})^2 \geq 0 \therefore 1 - 2\sqrt{a_i} + a_i \geq 0$$

$$\therefore 1 + a_i \geq 2\sqrt{a_i}$$

$$\prod_{i=1}^n (1 + a_i) \geq \prod_{i=1}^n 2\sqrt{a_i} = 2^n \sqrt{a_1 a_2 \dots a_n}$$

$$\text{But } a_1 a_2 \dots a_n = 1 \text{ (given)}$$

$$\therefore (1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n$$

$$17. \quad 3(a^2 + b^2 + c^2)$$

$$= a^2 + b^2 + c^2 + (a^2 + b^2) + (b^2 + c^2) + (c^2 + a^2)$$

$$\geq a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$= (a + b + c)^2$$

$$18. \quad x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$$

$$(a) \quad a^2 + ab + b^2 = \left(a + \frac{b}{2}\right)^2 + \frac{3}{4}b^2 > 0$$

$$(b) \quad a^4 + b^4 - a^3b - ab^3$$

$$= a^3(a - b) + b^3(b - a)$$

$$= (a - b)(a^3 - b^3)$$

$$= (a - b)(a - b)(a^2 + ab + b^2)$$

$$= (a - b)^2(a^2 + ab + b^2) > 0$$

$$19. \quad \text{Given } x^2 + y^2 + z^2 \geq xy + yz + zx$$

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$$

$$\geq xy + yz + zx + 2xy + 2yz + 2zx$$

$$= 3(xy + yz + zx)$$

20 over
the page

$$20. \quad x_0 > x_1 > x_2 > \dots > x_n$$

$$x_0 - x_n = x_0 - x_1 + x_1 - x_2 + x_2 - x_3 + \dots + x_{n-1} - x_n$$

$$x_0 - x_1 + \frac{1}{x_0 - x_1} \geq 2 \quad \text{and} \quad x_i - x_{i+1} + \frac{1}{x_i - x_{i+1}} \geq 2, \quad i=0, \dots, n-1$$

$$x_0 - x_n + \frac{1}{x_0 - x_1} + \frac{1}{x_1 - x_2} + \dots + \frac{1}{x_{n-1} - x_n} \geq \underbrace{2 + \dots + 2}_{n \text{ times}} = 2n$$

Hence the result.

20. $a^2b^2 + b^2c^2 + c^2a^2 \geq abc(a+b+c)$

$$a^2b^2 + b^2c^2 \geq 2ab^2c$$

$$a^2b^2 + c^2a^2 \geq 2a^2bc$$

$$b^2c^2 + c^2a^2 \geq 2abc^2$$

$$2(a^2b^2 + b^2c^2 + c^2a^2) \geq 2(ab^2c + a^2bc + abc^2)$$

$$a^2b^2 + b^2c^2 + c^2a^2 \geq abc(a+b+c)$$

$$\leq \frac{1}{2} (a+b+c)^2 = \frac{1}{2} (a+b+c)^2$$

$$(a+b+c)^2 \geq 4abc$$

$$(a+b+c)^2 \geq 4abc$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \geq 4abc$$

$$a^2 + b^2 + c^2 \geq 4abc - 2ab - 2bc - 2ca$$

$$a^2 + b^2 + c^2 \geq 4abc - 2ab - 2bc - 2ca$$

$$(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$$

$$(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$$

$$(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$$

$$(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \geq 4abc$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \geq 4abc$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \geq 4abc$$

$$(a+b+c)^2 \geq 4abc$$

$$(a+b+c)^2 \geq 4abc$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \geq 4abc$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \geq 4abc$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \geq 4abc$$

Have the result.

$$22. \quad a^2 + b^2 \geq 2ab$$

$$a^2 + b^2 = (a+b)^2 - 2ab \geq 2ab$$

$$(a+b)^2 \geq 4ab$$

$$ab \leq \frac{(a+b)^2}{4} = \frac{1}{4}$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$1 \leq a^2 + b^2 + \frac{1}{2}$$

$$\therefore a^2 + b^2 \geq \frac{1}{2}$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$= a^2 - ab + b^2$$

$$= a^2 + b^2 - ab$$

$$\geq \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$23. \quad a^4 + b^4 = (a^2 + b^2)(a^2 + b^2) - 2a^2b^2$$

$$\geq 2ab(a^2 + b^2) - 2a^2b^2$$

$$= ab(2a^2 + 2b^2 - 2ab)$$

$$= ab(a^2 + b^2 + a^2 + b^2 - 2ab)$$

$$= ab(a^2 + b^2 + (a-b)^2)$$

$$\geq ab(a^2 + b^2)$$

$$\text{since } a^2 + b^2 + (a-b)^2 \geq a^2 + b^2$$

5. Application of the AM-GM inequality with three numbers

$$\begin{aligned}
 1. \quad \frac{1}{3}(a^3-b^3)(a-b) &\geq ab(a-b)^2 \\
 \frac{1}{3}(a^3-b^3)(a-b) &= \frac{1}{3}(a-b)^2(a^2+ab+b^2) \\
 &= (a-b)^2 \left(\frac{a^2+ab+b^2}{3} \right) \\
 &\geq (a-b)^2 \sqrt[3]{a^2abb^2} \\
 &= (a-b)^2 ab.
 \end{aligned}$$

$$2. \text{ Given } abc=1, \text{ prove } a^2+b^2+c^2 \geq a+b+c$$

Method 2
on the back
page

First prove that $3(a^2+b^2+c^2) \geq (a+b+c)^2$ (see question 17/section 3)

$$\begin{aligned}
 3(a^2+b^2+c^2) &\geq (a+b+c)^2 \\
 &= (a+b+c) \times (a+b+c) \\
 &\geq (a+b+c) \times 3\sqrt[3]{abc} = 3(a+b+c)
 \end{aligned}$$

$$\therefore a^2+b^2+c^2 \geq a+b+c.$$

$$3. \quad \frac{a^3}{b} + \frac{b^3}{c} + bc \geq 3ab$$

$$\frac{a^3}{b} + \frac{c^3}{a} + ab \geq 3ac$$

$$\frac{b^3}{c} + \frac{c^3}{a} + ac \geq 3bc$$

$$2 \left(\frac{a^3}{b} + \frac{b^3}{c} + \frac{c^3}{a} \right) \geq 2(ab+bc+ca)$$

$$\therefore \frac{a^3}{b} + \frac{b^3}{c} + \frac{c^3}{a} \geq ab+bc+ca.$$

$$4. \quad \begin{aligned} a^2b+b^2c+c^2a &\geq 3\sqrt[3]{a^2b \cdot b^2c \cdot c^2a} = 3abc \\ ab^2+bc^2+ca^2 &\geq 3\sqrt[3]{ab^2 \cdot bc^2 \cdot ca^2} = 3abc \end{aligned}$$

$$\begin{aligned}
 (a^2b+b^2c+c^2a)(ab^2+bc^2+ca^2) &\geq (3abc)(3abc) \\
 &= 9a^2b^2c^2
 \end{aligned}$$

$$5. \quad \frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$$

$$\begin{aligned} 2\left(\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b}\right) &= 2\left(\frac{a+b+c}{b+c} + \frac{a+b+c}{a+c} + \frac{a+b+c}{a+b}\right) - 6 \\ &= 2(a+b+c)\left(\frac{1}{b+c} + \frac{1}{a+c} + \frac{1}{a+b}\right) - 6 \\ &= (a+b+b+c+c+a)\left(\frac{1}{b+c} + \frac{1}{a+c} + \frac{1}{a+b}\right) - 6 \\ &\geq 3\sqrt{(a+b)(b+c)(c+a)} \times 3\sqrt{\frac{1}{b+c} \times \frac{1}{a+c} \times \frac{1}{a+b}} - 6 \\ &\geq 9 - 6 = 3 \end{aligned}$$

$$\therefore \frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}.$$

$$6. \quad abc = 1$$

$$\begin{aligned} &\frac{1+ab}{1+a} + \frac{1+bc}{1+b} + \frac{1+ac}{1+c} \\ &= \frac{abc+ab}{1+a} + \frac{abc+bc}{1+b} + \frac{abc+ac}{1+c} \\ &= \frac{ab(1+c)}{1+a} + \frac{bc(1+a)}{1+b} + \frac{ac(1+b)}{1+c} \\ &\geq 3\sqrt{\frac{ab(1+c)}{1+a} \times \frac{bc(1+a)}{1+b} \times \frac{ac(1+b)}{1+c}} \\ &= 3\sqrt{a^2b^2c^2} = 3. \end{aligned}$$

7. see section 3 question 15.

$$(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 3\sqrt{abc} \times 3\sqrt{\frac{1}{a} \frac{1}{b} \frac{1}{c}} = 9$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a+b+c}$$

$$2(a+b+c)\left(\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a}\right) = (a+b+b+c+c+a)\left(\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a}\right) \geq 9$$

(see previous).

$$8. \frac{1}{b(a+b)} + \frac{1}{c(b+c)} + \frac{1}{a(c+a)} \geq \frac{27}{2(a+b+c)^2}$$

$$\begin{aligned} & 2(a+b+c)^2 \left(\frac{1}{b(a+b)} + \frac{1}{c(b+c)} + \frac{1}{a(c+a)} \right) \\ &= (a+b+c)(a+b+b+c+c+a) \left(\frac{1}{b(a+b)} + \frac{1}{c(b+c)} + \frac{1}{a(c+a)} \right) \\ &\geq \sqrt[3]{abc} \times 3 \sqrt[3]{(a+b)(b+c)(c+a)} \times 3 \sqrt[3]{\frac{1}{b(a+b)} \times \frac{1}{c(b+c)} \times \frac{1}{a(c+a)}} \\ &= 27 \end{aligned}$$

$$\therefore \frac{1}{b(a+b)} + \frac{1}{c(b+c)} + \frac{1}{a(c+a)} \geq \frac{27}{2(a+b+c)^2}$$

$$9. \quad abc = 1$$

$$\frac{a}{(a+1)(b+1)} + \frac{b}{(b+1)(c+1)} + \frac{c}{(c+1)(a+1)} \geq \frac{3}{4}$$

Multiply both sides by $4(a+1)(b+1)(c+1)$

$$4(a(c+1) + b(a+1) + c(b+1)) \geq 3(a+1)(b+1)(c+1)$$

which is equivalent to

$$4(ab+bc+ac+a+b+c) \geq 3(abc+ab+bc+ca+a+b+c+1)$$

or

$$ab+bc+ac+a+b+c \geq 3(abc+1) = 3(1+1) = 6$$

Alternatively: $\frac{a}{(a+1)(b+1)} + \frac{b}{(b+1)(c+1)} + \frac{c}{(c+1)(a+1)}$

$$= 1 - \frac{2}{(a+1)(b+1)(c+1)}$$

But $(a+1)(b+1)(c+1) \geq 8 \therefore \frac{-2}{(a+1)(b+1)(c+1)} \geq -\frac{1}{4}$

$$\therefore \frac{a}{(a+1)(b+1)} + \frac{b}{(b+1)(c+1)} + \frac{c}{(c+1)(a+1)} \geq 1 - \frac{1}{4} = \frac{3}{4}$$

10. Given that $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = 1$. $abc \geq 8$?

$$(a+1+b+1+c+1) \left(\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} \right) \geq 9$$

$$a+b+c+3 \geq 9 \quad \therefore a+b+c \geq 6$$

$$\boxed{a+b+c+2 \geq 8} \quad (1)$$

But $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = 1$

$$\therefore (b+1)(c+1) + (a+1)(c+1) + (a+1)(b+1) = (a+1)(b+1)(c+1)$$

$$\therefore bc+b+c+1+ac+a+c+1+ab+a+b+1 = abc+ab+bc+ca+a+b+c+1$$

$$\boxed{a+b+c+2 = abc} \quad (2)$$

Combining (1) and (2), we obtain $abc \geq 8$.

11. $a^2+b^2+c^2 = ab+bc+ca$

$$\therefore 2(a^2+b^2+c^2) = 2(ab+bc+ca)$$

$$\therefore (a^2+b^2-2ab) + (a^2+c^2-2ac) + (b^2+c^2-2bc) = 0$$

$$\therefore (a-b)^2 + (a-c)^2 + (b-c)^2 = 0$$

$$\therefore a-b=0, a-c=0 \text{ and } b-c=0$$

$$\therefore a=b, a=c \text{ and } b=c$$

$$\therefore a=b=c$$

and the triangle $\triangle ABC$ is an equilateral triangle.

6. Mixed problems:

1. $m, n, p, q > 0$

(a) $(\sqrt{m} - \sqrt{n})^2 \geq 0$

$$\therefore m - 2\sqrt{mn} + n \geq 0$$

$$\therefore m + n \geq 2\sqrt{mn}$$

(b) $m + n \geq 2\sqrt{mn}$
 $n + p \geq 2\sqrt{np}$
 $m + p \geq 2\sqrt{mp}$

$$(m+n)(n+p)(p+m) \geq 8\sqrt{mnnpmp} = 8mnp$$

(c) $\frac{m}{n} + \frac{n}{p} + \frac{p}{q} + \frac{q}{m} \geq 4$

$$\frac{m}{n} + \frac{n}{p} \geq 2\sqrt{\frac{m}{n} \times \frac{n}{p}} = 2\sqrt{\frac{m}{p}}$$

$$\frac{p}{q} + \frac{q}{m} \geq 2\sqrt{\frac{p}{q} \times \frac{q}{m}} = 2\sqrt{\frac{p}{m}}$$

$$\begin{aligned} \frac{m}{n} + \frac{n}{p} + \frac{p}{q} + \frac{q}{m} &\geq 2\left(\sqrt{\frac{m}{p}} + \sqrt{\frac{p}{m}}\right) \\ &\geq 2\left(2\sqrt{\frac{m}{p} \times \frac{p}{m}}\right) \\ &= 4. \end{aligned}$$

2. $a > 0, b > 0, a^3 + b^3 \geq a^2b + ab^2$

$$(a-b)^2 \geq 0 \therefore a^2 - 2ab + b^2 \geq 0$$

$$a^2 - ab + b^2 \geq ab$$

$$\begin{aligned} a^3 + b^3 &= (a+b)(a^2 - ab + b^2) \\ &\geq (a+b)ab \\ &= a^2b + ab^2. \end{aligned}$$

3. $a, b, c > 0$

(a) $a^2 + b^2 + b^2 + c^2 + c^2 + a^2 \geq 2ab + 2bc + 2ac$
 $\therefore a^2 + b^2 + c^2 \geq ab + bc + ac$

(b) $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq \sqrt[3]{abc} \times \sqrt[3]{\frac{1}{abc}}$
 $= 9$

$$\therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a+b+c}$$

$$(c) \quad a^2 + b^2 + c^2 = 9$$

$$\frac{1}{1+ab} + \frac{1}{1+bc} + \frac{1}{1+ac} \geq \frac{3}{4}$$

$$\begin{aligned} & (1+ab+1+bc+1+ac) \left(\frac{1}{1+ab} + \frac{1}{1+bc} + \frac{1}{1+ac} \right) \\ & \geq 3 \sqrt[3]{(1+ab)(1+bc)(1+ac)} \times 3 \sqrt[3]{\frac{1}{1+ab} \times \frac{1}{1+bc} \times \frac{1}{1+ac}} \\ & = 9 \end{aligned}$$

$$\frac{1}{1+ab} + \frac{1}{1+bc} + \frac{1}{1+ac} \geq \frac{9}{3+ab+bc+ca}$$

$$\begin{aligned} \text{from (a)} \quad a^2 + b^2 + c^2 & \geq ab + bc + ca \\ 9 & \geq ab + bc + ca \\ 12 & \geq 3 + ab + bc + ca \\ \therefore \frac{1}{3+ab+bc+ca} & \geq \frac{1}{12} \end{aligned}$$

$$\therefore \frac{1}{1+ab} + \frac{1}{1+bc} + \frac{1}{1+ac} \geq 9 \times \frac{1}{12} = \frac{3}{4}$$

$$4. (a) \quad (a-3b)^2 \geq 0 \therefore a^2 - 6ab + 9b^2 \geq 0 \\ \therefore a^2 + 9b^2 \geq 6ab$$

$$\begin{aligned} (b) \quad a^2 + 9b^2 & \geq 6ab \\ a^2 + 9c^2 & \geq 6ac \\ b^2 + 9c^2 & \geq 6bc \end{aligned}$$

$$\begin{aligned} 2a^2 + 10b^2 + 18c^2 & \geq 6(ab + bc + ac) \\ a^2 + 5b^2 + 9c^2 & \geq 3(ab + bc + ca) \end{aligned}$$

$$(c) \quad a > b > c \therefore ab > ac > bc$$

$$\therefore a^2 + 5b^2 + 9c^2 \geq 3(bc + bc + bc) = 9bc$$

5. $a+b+c=1$ and $a+b+c \geq \sqrt[3]{abc}$

(a) let $a = \frac{1}{xy}$, $b=x$ and $c=y$

$$\frac{1}{xy} + x + y \geq \sqrt[3]{\frac{1}{xy} \cdot xy} = 3$$

(b) $\frac{1}{a(a+1)} + a + a+1 \geq 3$

$$\frac{1}{b(b+1)} + b + b+1 \geq 3$$

$$\frac{1}{c(c+1)} + c + c+1 \geq 3$$

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} + 2(a+b+c) + 3 \geq 9$$

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} \geq 9 - 5 = 4$$

6. (a) $(a-b)^2 \geq 0 \therefore a^2 - 2ab + b^2 \geq 0$

$$\therefore a^2 + b^2 \geq 2ab$$

(b) $\left(\sqrt{\frac{x}{y}} - \sqrt{\frac{y}{x}}\right)^2 \geq 0 \therefore \frac{x}{y} - 2\sqrt{\frac{xy}{yx}} + \frac{y}{x} \geq 0$

$$\therefore \frac{x}{y} + \frac{y}{x} \geq 2$$

(c) prove by induction that

$$(x_1 + \dots + x_n) \left(\frac{1}{x_1} + \dots + \frac{1}{x_n} \right) \geq n^2$$

For $n=1$, $x_1 \times \frac{1}{x_1} = 1 \geq 1^2$ true for $n=1$

Assume it is true for $n=k$

$$(x_1 + \dots + x_k) \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} \right) \geq k^2 \quad (*)$$

show it is true for $n=k+1$.

$$\begin{aligned} & (x_1 + \dots + x_k + x_{k+1}) \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} + \frac{1}{x_{k+1}} \right) \\ &= (x_1 + \dots + x_k) \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} \right) + (x_1 + \dots + x_k + x_{k+1}) \left(\frac{1}{x_{k+1}} \right) \\ &\geq k^2 + \left(\frac{x_1}{x_{k+1}} + \frac{x_{k+1}}{x_1} \right) + \left(\frac{x_2}{x_{k+1}} + \frac{x_{k+1}}{x_2} \right) + \dots + \left(\frac{x_k}{x_{k+1}} + \frac{x_{k+1}}{x_k} \right) + \frac{x_{k+1}}{x_{k+1}} \\ &\geq k^2 + \underbrace{2 + \dots + 2}_{k \text{ times}} + 1 = k^2 + 2k + 1 = (k+1)^2 \end{aligned}$$

Hence By Mathematical induction it is true for all $n \geq 1$.

$$7. (a^3+2)(b^3+2)(c^3+2) \geq (a+b+c)^3$$

$$\frac{a^3}{a^3+2} + \frac{1}{b^3+2} + \frac{1}{c^3+2} \geq \frac{3a}{\sqrt[3]{(a^3+2)(b^3+2)(c^3+2)}}$$

$$\frac{1}{a^3+2} + \frac{b^3}{b^3+2} + \frac{1}{c^3+2} \geq \frac{3b}{\sqrt[3]{(a^3+2)(b^3+2)(c^3+2)}}$$

$$\frac{1}{a^3+2} + \frac{1}{b^3+2} + \frac{c^3}{c^3+2} \geq \frac{3c}{\sqrt[3]{(a^3+2)(b^3+2)(c^3+2)}}$$

$$\frac{a^3+2}{a^3+2} + \frac{b^3+2}{b^3+2} + \frac{c^3+2}{c^3+2} \geq \frac{3(a+b+c)}{\sqrt[3]{(a^3+2)(b^3+2)(c^3+2)}}$$

$$1 \geq \frac{a+b+c}{\sqrt[3]{(a^3+2)(b^3+2)(c^3+2)}}$$

$$\therefore (a^3+2)(b^3+2)(c^3+2) \geq (a+b+c)^3$$

8. (Jomaa 2019)

$$1+a = a+a+b+c \\ = a+b+a+c \geq 2\sqrt{(a+b)(a+c)}$$

$$1+b = a+b+b+c \geq 2\sqrt{(a+b)(b+c)}$$

$$1+c = a+c+b+c \geq 2\sqrt{(b+c)(a+c)}$$

$$\frac{2}{1+a} \leq \frac{1}{\sqrt{(a+b)(a+c)}} \therefore \frac{4}{(1+a)^2} \leq \frac{1}{(a+b)(a+c)}$$

$$\text{Similarly } \frac{4}{(1+b)^2} \leq \frac{1}{(a+b)(b+c)}$$

$$\text{and } \frac{4}{(1+c)^2} \leq \frac{1}{(a+c)(b+c)}$$

$$\begin{aligned}
\frac{4}{(a+1)^2} + \frac{4}{(b+1)^2} + \frac{4}{(c+1)^2} &\leq \frac{1}{(a+b)(a+c)} + \frac{1}{(a+b)(b+c)} + \frac{1}{(a+c)(b+c)} \\
&= \frac{b+c+a+c+a+b}{(a+b)(b+c)(c+a)} \\
&= \frac{2}{(a+b)(b+c)(c+a)}
\end{aligned}$$

$$\therefore \frac{2}{(1+a)^2} + \frac{2}{(1+b)^2} + \frac{2}{(1+c)^2} \leq \frac{1}{(a+b)(b+c)(c+a)}$$

Q

(a) $(a_1 b_2 - a_2 b_1)^2 \geq 0$

$$a_1^2 b_2^2 - 2a_1 b_2 a_2 b_1 + a_2^2 b_1^2 \geq 0$$

$$a_1^2 b_2^2 + a_2^2 b_1^2 \geq 2a_1 a_2 b_1 b_2$$

$$a_1^2 b_2^2 + a_2^2 b_1^2 + a_1^2 b_1^2 + a_2^2 b_2^2 \geq a_1^2 b_1^2 + 2a_1 a_2 b_1 b_2 + a_2^2 b_2^2$$

$$a_1^2 (b_1^2 + b_2^2) + a_2^2 (b_1^2 + b_2^2) \geq (a_1 b_1 + a_2 b_2)^2$$

$$(a_1^2 + a_2^2)(b_1^2 + b_2^2) \geq (a_1 b_1 + a_2 b_2)^2$$

(b) prove by induction that

$$(a_1^2 + \dots + a_n^2)(b_1^2 + \dots + b_n^2) \geq (a_1 b_1 + \dots + a_n b_n)^2$$

for $n=1$, $a_1^2 b_1^2 \geq (a_1 b_1)^2 = a_1^2 b_1^2$

Assume it is true for $n=k$

$$(a_1^2 + \dots + a_k^2)(b_1^2 + \dots + b_k^2) \geq (a_1 b_1 + \dots + a_k b_k)^2 \quad (*)$$

prove it true for $n=k+1$

$$\text{i.e. } (a_1^2 + \dots + a_k^2 + a_{k+1}^2)(b_1^2 + \dots + b_k^2 + b_{k+1}^2) \geq (a_1 b_1 + \dots + a_k b_k + a_{k+1} b_{k+1})^2$$

$$(a_1^2 + \dots + a_k^2 + a_{k+1}^2)(b_1^2 + \dots + b_k^2 + b_{k+1}^2)$$

$$= (a_1^2 + \dots + a_k^2)(b_1^2 + \dots + b_k^2) + (a_1^2 + \dots + a_k^2)b_{k+1}^2 + a_{k+1}^2(b_1^2 + \dots + b_k^2) + a_{k+1}^2 b_{k+1}^2$$

$$\geq (a_1 b_1 + \dots + a_k b_k)^2 + a_1^2 b_{k+1}^2 + a_{k+1}^2 b_1^2 + \dots + a_k^2 b_{k+1}^2 + a_{k+1}^2 b_k^2 + a_{k+1}^2 b_{k+1}^2 \quad (\text{using } *)$$

$$\geq (a_1 b_1 + \dots + a_k b_k)^2 + 2a_1 b_1 a_{k+1} b_{k+1} + \dots + 2a_k b_k a_{k+1} b_{k+1} + a_{k+1}^2 b_{k+1}^2$$

$$= (a_1 b_1 + \dots + a_k b_k)^2 + 2a_{k+1} b_{k+1} (a_1 b_1 + \dots + a_k b_k) + a_{k+1}^2 b_{k+1}^2$$

$$= (a_1 b_1 + \dots + a_k b_k + a_{k+1} b_{k+1})^2$$

$\therefore$ Hence by mathematical induction, it is true for all $n \geq 1$.

$$(c) \quad \frac{a^2}{y+z} + \frac{b^2}{x+z} + \frac{c^2}{x+y} \geq \frac{a+b+c}{2}$$

$$2(x+y+z) \left[\frac{a^2}{y+z} + \frac{b^2}{x+z} + \frac{c^2}{x+y} \right]$$

$$= (y+z+x+z+x+y) \left(\frac{a^2}{y+z} + \frac{b^2}{x+z} + \frac{c^2}{x+y} \right)$$

$$\geq (a+b+c)^2$$

$$\text{But } x+y+z = a+b+c$$

$$\therefore \frac{a^2}{y+z} + \frac{b^2}{x+z} + \frac{c^2}{x+y} \geq \frac{a+b+c}{2}$$

10 (NSBHS, Mar 19):

(a) $(a-b)^2 \geq 0 \therefore a^2 - 2ab + b^2 \geq 0 \therefore a^2 + b^2 \geq 2ab$

(b) $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$

from (a) $a^2 + b^2 \geq 2ab$

similarly $b^2 + c^2 \geq 2bc$

$c^2 + a^2 \geq 2ac$

$$a^2 + b^2 + b^2 + c^2 + c^2 + a^2 \geq 2ab + 2bc + 2ac$$

$$\therefore 2(a^2 + b^2 + c^2) \geq 2(ab + bc + ac)$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ac$$

$$\therefore a^2 + b^2 + c^2 - ab - bc - ac \geq 0$$

but $a+b+c \geq 0$

$$\therefore (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac) \geq 0$$

$$\therefore a^3 + b^3 + c^3 - 3abc \geq 0$$

$$\therefore \frac{a^3 + b^3 + c^3}{3} \geq abc$$

(c) From part (b), let $c=1$

$$\frac{a^3 + b^3 + 1^3}{3} \geq ab(1)$$

$$\therefore \frac{a^3 + b^3 + 1}{3} \geq ab$$

(d) From (c) $\frac{a^3 + b^3 + 1}{3} \geq ab$

Similarly $\frac{a^3 + c^3 + 1}{3} \geq ac$

and $\frac{b^3 + c^3 + 1}{3} \geq bc$

From (b) $\frac{a^3 + b^3 + c^3}{3} \geq abc$

$$\frac{a^3 + b^3 + c^3}{3} + \frac{a^3 + b^3 + 1}{3} + \frac{a^3 + c^3 + 1}{3} + \frac{b^3 + c^3 + 1}{3} \geq abc + ab + ac + bc$$
$$\frac{3(a^3 + b^3 + c^3 + 1)}{3} \geq abc + ab + ac + bc \therefore a^3 + b^3 + c^3 + 1 \geq abc + ab + bc + ca$$

11. (a) $(a-b) \geq 0 \therefore a^2 - 2ab + b^2 \geq 0 \therefore a^2 + b^2 \geq 2ab$

(b) $(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
 $= a^3 + ab^2 + ac^2 - a^2b - abc - ca^2 +$
 $ba^2 + b^3 + bc^2 - ab^2 - b^2c - abc +$
 $ca^2 + cb^2 + c^3 - abc - bc^2 - c^2a$
 $= a^3 + b^3 + c^3 - 3abc$

(c) use (a) $\therefore a^2 + b^2 \geq 2ab$
 similarly $a^2 + c^2 \geq 2ac$
 $b^2 + c^2 \geq 2bc$

$$a^2 + b^2 + b^2 + c^2 + c^2 + a^2 \geq 2ab + 2bc + 2ac$$

$$\therefore 2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca$$

(d) From (c) $a^2 + b^2 + c^2 \geq ab + bc + ca$
 $\therefore a^2 + b^2 + c^2 - ab - bc - ca \geq 0$
 and $a+b+c \geq 0$

use (b) $\therefore a^3 + b^3 + c^3 - 3abc \geq 0$
 $\therefore a^3 + b^3 + c^3 \geq 3abc$

(e) $2(a^3 + b^3 + c^3) + \frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3}$
 $= a^3 + b^3 + c^3 + (a^3 + b^3 + c^3) + \frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3}$
 $= a^3 + b^3 + c^3 + \left(a^3 + \frac{1}{a^3}\right) + \left(b^3 + \frac{1}{b^3}\right) + \left(c^3 + \frac{1}{c^3}\right)$

use (a) for a^3 and $\frac{1}{a^3} \therefore a^3 + \frac{1}{a^3} \geq 2\sqrt{a^3 \times \frac{1}{a^3}} = 2$
 similarly $b^3 + \frac{1}{b^3} \geq 2$ and $c^3 + \frac{1}{c^3} \geq 2$

use (d) $a^3 + b^3 + c^3 \geq 3abc$

Now
 $2(a^3 + b^3 + c^3) + \frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} = a^3 + b^3 + c^3 + \left(a^3 + \frac{1}{a^3}\right) + \left(b^3 + \frac{1}{b^3}\right) + \left(c^3 + \frac{1}{c^3}\right)$
 $\geq 3abc + 2 + 2 + 2$
 $= 3abc + 6$

$$(f) \quad a+b+c+a^2+b^2+c^2+a^3+b^3+c^3 \geq 3(ab+bc+ca)$$

$$a+b+c+a^3+b^3+c^3 = a+a^3+b+b^3+c+c^3$$

$$a+a^3 \geq 2\sqrt{aa^3} = 2a^2$$

$$b+b^3 \geq 2b^2$$

$$c+c^3 \geq 2c^2$$

$$\therefore a+b+c+a^3+b^3+c^3 \geq 2(a^2+b^2+c^2)$$

$$\begin{aligned} \therefore a+b+c+a^2+b^2+c^2+a^3+b^3+c^3 &\geq 2(a^2+b^2+c^2) + a^2+b^2+c^2 \\ &= 3(a^2+b^2+c^2) \\ &\geq 3(ab+bc+ca) \end{aligned}$$

$$12. \quad \begin{aligned} a^3+b^3+a^3 &\geq 3a^2b \\ b^3+c^3+b^3 &\geq 3b^2c \\ a^3+c^3+c^3 &\geq 3ac^2 \end{aligned}$$

$$3a^3+3b^3+3c^3 \geq 3(a^2b+b^2c+c^2a)$$

$$\therefore a^3+b^3+c^3 \geq a^2b+b^2c+c^2a$$

$$(ab)^3+(ac)^3+(ab)^3 \geq 3a^3b^2c = 3a^2b \quad (\text{since } abc=1)$$

$$\text{Similarly } (bc)^3+(ac)^3+(ac)^3 \geq 3a^2bc^3 = 3ac^2 \quad (\text{since } abc=1)$$

$$\text{and } (bc)^3+(bc)^3+(ab)^3 \geq 3b^3c^2a = 3b^2c \quad (\text{since } abc=1)$$

$$\therefore 3(ab)^3+3(bc)^3+3(ca)^3 \geq 3(a^2b+b^2c+ac^2)$$

$$\therefore (ab)^3+(bc)^3+(ca)^3 \geq a^2b+b^2c+ac^2$$

$$\begin{aligned} \therefore a^3+b^3+c^3+(ab)^3+(bc)^3+(ca)^3 &\geq a^2b+b^2c+c^2a+a^2b+b^2c+ac^2 \\ &= 2(a^2b+b^2c+c^2a) \end{aligned}$$

13.

$$\frac{a^2}{b^2} + \frac{b^2}{c^2} \geq 2 \sqrt{\frac{a^2}{b^2} \times \frac{b^2}{c^2}} = \frac{2a}{c}$$

$$\frac{a^2}{b^2} + \frac{c^2}{a^2} \geq 2 \frac{c}{b}$$

$$\frac{b^2}{c^2} + \frac{c^2}{a^2} \geq 2 \frac{b}{a}$$

$$2 \frac{a^2}{b^2} + 2 \frac{b^2}{c^2} + 2 \frac{c^2}{a^2} \geq 2 \frac{b}{a} + 2 \frac{a}{c} + 2 \frac{c}{b}$$

$$\therefore \frac{a^2}{b^2} + \frac{b^2}{c^2} + \frac{c^2}{a^2} \geq \frac{b}{a} + \frac{c}{b} + \frac{a}{c}.$$

14.

$$\frac{1}{a^2} + \frac{1}{b^2} \geq 2 \sqrt{\frac{1}{a^2} \frac{1}{b^2}} = \frac{2}{ab} = \frac{2c}{abc}$$

$$\frac{1}{a^2} + \frac{1}{c^2} \geq \frac{2}{ac} = \frac{2b}{abc}$$

$$\frac{1}{b^2} + \frac{1}{c^2} \geq \frac{2}{bc} = \frac{2a}{abc}$$

$$\frac{2}{a^2} + \frac{2}{b^2} + \frac{2}{c^2} \geq \frac{2a}{abc} + \frac{2b}{abc} + \frac{2c}{abc}$$

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{a+b+c}{abc}.$$

$$15. \textcircled{a} \quad (a^2+1)(b^2+1) = a^2b^2 + a^2 + b^2 + 1$$

$$a^2b^2 + 1 \geq 2\sqrt{a^2b^2} = 2ab$$

$$\begin{aligned} (a^2+1)(b^2+1) &= a^2b^2 + a^2 + b^2 + 1 \\ &\geq a^2 + b^2 + 2ab \\ &= (a+b)^2. \end{aligned}$$

$$\textcircled{b} \quad (a^2+1)(b^2+1) = a^2b^2 + a^2 + b^2 + 1$$

$$a^2 + b^2 \geq 2ab$$

$$\begin{aligned} (a^2+1)(b^2+1) &= a^2b^2 + a^2 + b^2 + 1 \\ &\geq a^2b^2 + 2ab + 1 \\ &= (ab+1)^2. \end{aligned}$$

$$\textcircled{c} \quad (a^2+b^2)(c^2+d^2) = a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$$

$$a^2d^2 + b^2c^2 \geq 2\sqrt{a^2d^2b^2c^2} = 2abcd$$

$$\begin{aligned} (a^2+b^2)(c^2+d^2) &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 \\ &\geq a^2c^2 + b^2d^2 + 2abcd \\ &= (ac+bd)^2. \end{aligned}$$

$\textcircled{d}$

$$\begin{aligned} &(a^2+b^2+c^2)(x^2+y^2+z^2) \\ &= (a^2+b^2)(x^2+y^2) + z^2(a^2+b^2) + c^2(x^2+y^2) + c^2z^2 \\ &= (a^2+b^2)(x^2+y^2) + (a^2z^2+c^2x^2) + (b^2z^2+c^2y^2) + c^2z^2 \\ &\geq (ax+by)^2 + 2aczx + 2bczy + c^2z^2 \\ &= (ax+by)^2 + 2cz(ax+by) + c^2z^2 \\ &= (ax+by+cz)^2. \end{aligned}$$

HSC inequalities questions

1. (2015, ISC)

(a) Given that $\frac{x+y}{2} \geq \sqrt{xy}$ (1)

Sub x^v for x and y^v for y in (1) we obtain

$$\frac{x^v + y^v}{2} \geq \sqrt{x^v y^v} = xy$$

Take the square root of both sides, we obtain

$$\sqrt{\frac{x^v + y^v}{2}} \geq \sqrt{xy}$$

(b) use (a) for a and b , we obtain

$$\sqrt{\frac{a^2 + b^2}{2}} \geq \sqrt{ab} \quad \therefore \frac{a^2 + b^2}{2} \geq ab$$

Similarly $\sqrt{\frac{c^2 + d^2}{2}} \geq \sqrt{cd} \quad \therefore \frac{c^2 + d^2}{2} \geq cd$

$$\begin{aligned} \frac{a^2 + b^2 + c^2 + d^2}{4} &= \frac{1}{2} \left(\frac{a^2 + b^2}{2} + \frac{c^2 + d^2}{2} \right) \\ &\geq \frac{1}{2} (ab + cd) \\ &\geq \sqrt{abcd} \quad \text{using (part a)} \end{aligned}$$

$$\therefore \sqrt{\frac{a^2 + b^2 + c^2 + d^2}{4}} \geq \sqrt{\sqrt{abcd}} = \sqrt[4]{abcd}$$

2. (2014, 15a)

$$a+b+c=1$$

$$\therefore (a+b+c)^2=1$$

$$\therefore 1 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

$$a \leq b \therefore a^2 \leq ab$$

$$a \leq c \therefore a^2 \leq ac$$

$$b \leq c \therefore b^2 \leq bc$$

$$1 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

$$\geq a^2 + b^2 + c^2 + 2a^2 + 2a^2 + 2b^2$$

$$= 5a^2 + 3b^2 + c^2$$

3. (2012, 15a)

$$(a) \quad (\sqrt{a} - \sqrt{b})^2 \geq 0 \therefore a - 2\sqrt{ab} + b \geq 0$$

$$\therefore a + b \geq 2\sqrt{ab}$$

$$\therefore \frac{a+b}{2} \geq \sqrt{ab}$$

(b)

$$1 \leq x \leq y \therefore x-1 \geq 0$$

$$\text{and } y-x \geq 0$$

$$\therefore (x-1)(y-x) \geq 0$$

$$xy - x^2 - y + x \geq 0$$

$$xy - x^2 + x \geq y$$

$$x(y-x+1) \geq y$$

$$(c) \quad 1 \leq j \leq n$$

$$\text{from (b)} \quad \therefore j(n-j+1) \geq n$$

$$\therefore \sqrt{j(n-j+1)} \geq \sqrt{n}$$

$$\frac{n+1}{2} \geq \sqrt{n+1} = \sqrt{n} \quad \text{from (a)}$$

$$\frac{n+1}{2} = \frac{(n-j+1)+j}{2} \geq \sqrt{j(n-j+1)}$$

(d)

$$\frac{n+1}{2} \geq \sqrt{n} \quad \therefore \left(\frac{n+1}{2}\right)^n \geq (\sqrt{n})^n$$

using $\frac{n+1}{2} \geq \sqrt{j(n-j+1)}$ for $j=1, \dots, n$, we obtain

$$\frac{n+1}{2} \geq \sqrt{1(n-1+1)} = \sqrt{n}$$

$$\frac{n+1}{2} \geq \sqrt{2(n-2+1)} = \sqrt{2(n-1)}$$

,

,

,

$$\frac{n+1}{2} \geq \sqrt{n(n-n+1)} = \sqrt{n}$$

$$\left(\frac{n+1}{2}\right)^n \geq \sqrt{1 \cdot n \times 2 \cdot (n-1) \times 3 \cdot (n-2) \times \dots \times n \cdot 1} = \sqrt{(n!)^2} = n!$$

$$\text{use (c)} \quad \sqrt{j(n-j+1)} \geq \sqrt{n} \quad \text{for } j=1, \dots, n$$

$$\prod_{j=1}^n \sqrt{j(n-j+1)} \geq \prod_{j=1}^n \sqrt{n}$$

$$n! \geq (\sqrt{n})^n$$

4. (2011, 5b) $p+q \geq r$

$$\begin{aligned} & \frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \\ &= \frac{p(1+q)(1+r) + q(1+p)(1+r) - r(1+p)(1+q)}{(1+p)(1+q)(1+r)} \\ &= \frac{p(1+q+r+qr) + q(1+p+r+pr) - r(1+p+q+pq)}{(1+p)(1+q)(1+r)} \\ &= \frac{p+pq+pr+pqr+q+qp+qr+pqr-r-rp-rq-rpq}{(1+p)(1+q)(1+r)} \\ &= \frac{p+q-r+2pq+pqr}{(1+p)(1+q)(1+r)} \end{aligned}$$

Give p, q, r all positive and $p+q-r \geq 0$

$$\therefore \frac{p+q-r+2pq+pqr}{(1+p)(1+q)(1+r)} \geq 0$$

$$\therefore \frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \geq 0.$$

5. (2004, 7a)

$$(a) \left(\sqrt{a} - \frac{1}{\sqrt{a}} \right)^2 \geq 0 \therefore a - 2 \times \sqrt{a} \times \frac{1}{\sqrt{a}} + \frac{1}{a} \geq 0$$

$$\therefore a + \frac{1}{a} \geq 2.$$

$$(b) (a_1 + \dots + a_n) \left(\frac{1}{a_1} + \dots + \frac{1}{a_n} \right) \geq n^2$$

prove by induction.

for $n=1$, $a_1 \times \frac{1}{a_1} \geq 1$

Assume it is true for $n=k$

$$(a_1 + a_2 + \dots + a_k) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} \right) \geq k^2 \quad (*)$$

prove it true for $n=k+1$

$$\text{i.e.: } (a_1 + \dots + a_k + a_{k+1}) \left(\frac{1}{a_1} + \dots + \frac{1}{a_k} + \frac{1}{a_{k+1}} \right) \geq (k+1)^2$$

$$(a_1 + \dots + a_k + a_{k+1}) \left(\frac{1}{a_1} + \dots + \frac{1}{a_k} + \frac{1}{a_{k+1}} \right)$$

$$= (a_1 + \dots + a_k) \left(\frac{1}{a_1} + \dots + \frac{1}{a_k} \right) +$$

$$(a_1 + \dots + a_k) \frac{1}{a_{k+1}} + a_{k+1} \left(\frac{1}{a_1} + \dots + \frac{1}{a_k} \right) + a_{k+1} \times \frac{1}{a_{k+1}}$$

$$\geq k^2 + \left(\frac{a_1}{a_{k+1}} + \frac{a_{k+1}}{a_1} \right) + \left(\frac{a_2}{a_{k+1}} + \frac{a_{k+1}}{a_2} \right) + \dots + \left(\frac{a_k}{a_{k+1}} + \frac{a_{k+1}}{a_k} \right) + 1$$

$$\geq k^2 + \underbrace{2 + 2 + \dots + 2}_k + 1 = k^2 + 2k + 1 = (k+1)^2$$

Hence by mathematical induction, it is true for all $n \geq 1$.

(c) use $(a_1 + a_2 + a_3) \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} \right) \geq 3^2$

$$(\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta) (\sin^2 \theta + \cos^2 \theta + \tan^2 \theta) \geq 9$$

$$(\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta) (1 + \tan^2 \theta) \geq 9$$

$$(\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta) \sec^2 \theta \geq 9$$

$$\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta \geq \frac{9}{\sec^2 \theta} = 9 \cos^2 \theta$$

6. (2003, 6C)

(a) $(\sqrt{x} - \sqrt{y})^2 \geq 0 \therefore x - 2\sqrt{xy} + y \geq 0$
 $\frac{x+y}{2} \geq \sqrt{xy}$

(b) $a^4 + b^4 \geq 2\sqrt{a^4 b^4} = 2a^2 b^2$ (using (a))

similarly $a^4 + c^4 \geq 2a^2 c^2$

$b^4 + c^4 \geq 2b^2 c^2$

$a^4 + b^4 + a^4 + c^4 + b^4 + c^4 \geq 2a^2 b^2 + 2a^2 c^2 + 2b^2 c^2$

$2(a^4 + b^4 + c^4) \geq 2(a^2 b^2 + a^2 c^2 + b^2 c^2)$

$\therefore a^4 + b^4 + c^4 \geq a^2 b^2 + a^2 c^2 + b^2 c^2$

(c) $a^2 b^2 + a^2 c^2 \geq 2\sqrt{a^2 b^2 a^2 c^2} = 2a^2 bc$

similarly $a^2 b^2 + b^2 c^2 \geq 2b^2 ac$

$a^2 c^2 + b^2 c^2 \geq 2c^2 ab$

$a^2 b^2 + a^2 c^2 + a^2 b^2 + b^2 c^2 + a^2 c^2 + b^2 c^2 \geq 2a^2 bc + 2b^2 ac + 2c^2 ab$

$2(a^2 b^2 + b^2 c^2 + a^2 c^2) \geq 2(a^2 bc + b^2 ac + c^2 ab)$

$a^2 b^2 + b^2 c^2 + a^2 c^2 \geq a^2 bc + b^2 ac + c^2 ab$

(d) from (b) and (c) we obtain

$a^4 + b^4 + c^4 \geq a^2 b^2 + a^2 c^2 + b^2 c^2 \geq a^2 bc + b^2 ac + c^2 ab$

$= abc(a + b + c)$

$= abc d$

since $a + b + c = d$ (given).

7. (2000, 8a)

$$(a) \quad (a-b)^2 \geq 0 \therefore a^2 - 2ab + b^2 \geq 0 \\ \therefore a^2 + b^2 \geq 2ab$$

$$a^2 + b^2 \geq 2ab \\ \text{Similarly } a^2 + c^2 \geq 2ac \\ b^2 + c^2 \geq 2bc$$

$$\therefore a^2 + b^2 + a^2 + c^2 + b^2 + c^2 \geq 2ab + 2ac + 2bc \\ 2(a^2 + b^2 + c^2) \geq 2(ab + ac + bc) \\ \therefore a^2 + b^2 + c^2 \geq ab + ac + bc$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc \\ \geq ab + ac + bc + 2ab + 2ac + 2bc \\ = 3(ab + bc + ca).$$

(b)

In $\triangle ABC$

$$b \leq a + c$$

$$\therefore b - c \leq a$$

$$\therefore (b-c)^2 \leq a^2$$

$$\therefore (b-c)^2 + 4bc \leq a^2 + 4bc$$

$$\therefore (b+c)^2 \leq a^2 + 4bc$$

$$\text{Similarly } (a+c)^2 \leq b^2 + 4ac$$

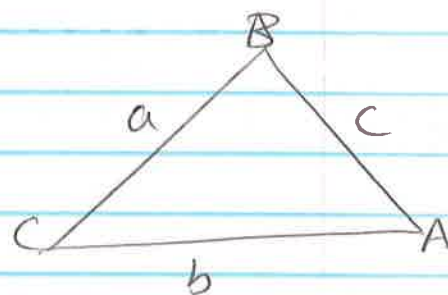
$$\text{and } (a+b)^2 \leq c^2 + 4ab$$

$$(a+b)^2 + (b+c)^2 + (c+a)^2 \leq c^2 + 4ab + a^2 + 4bc + b^2 + 4ac$$

$$a^2 + 2ab + b^2 + b^2 + 2bc + c^2 + c^2 + 2ac + a^2 \leq a^2 + b^2 + c^2 + 4(ab + bc + ca)$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ac + a^2 + b^2 + c^2 \leq a^2 + b^2 + c^2 + 4(ab + bc + ca)$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ac \leq 4(ab + bc + ca)$$



6. Trial exams inequality problems

1. NSBHS 2019

$$(a) \quad (a-bc)^2 \geq 0 \quad \therefore a^2 - 2abc + b^2c^2 \geq 0 \\ \therefore a^2 + (bc)^2 \geq 2abc$$

$$(b) \quad (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0 \\ \therefore a^2 - 2ab + b^2 + b^2 - 2bc + c^2 + c^2 - 2ca + a^2 \geq 0 \\ \therefore a^2 + b^2 + c^2 \geq ab + bc + ca$$

$$(c) \quad a^2(1+b^4) + b^2(1+c^4) + c^2(1+a^4) \\ = a^2 + a^2b^4 + b^2 + b^2c^4 + c^2 + a^4c^2 \\ = (a^2 + b^2c^4) + (b^2 + a^4c^2) + (c^2 + a^2b^4) \\ \geq 2abc + 2bac + 2cab \quad (\text{using (a)}) \\ = 6abc$$

$$(d) \quad a^2(1+a^4) + b^2(1+b^4) + c^2(1+c^4) \\ = a^2 + a^6 + b^2 + b^6 + c^2 + c^6 \\ = a^2 + b^2 + c^2 + a^4 + b^4 + c^4 \\ \geq a^2 + b^2 + c^2 + a^2b^2 + b^2c^2 + c^2a^2 \quad (\text{using (b)}) \\ \geq 6abc \quad (\text{using (c)})$$

2. NSGHS 2018, 15a

$$(a) \quad \left. \begin{array}{l} c \geq a \therefore (c-a) \geq 0 \\ d \geq b \therefore (d-b) \geq 0 \end{array} \right\} (c-a)(d-b) \geq 0$$

$\therefore$

$$cd - cb - ad + ab \geq 0 \\ ab + cd \geq bc + ad$$

$$(b) \quad x \geq y, \quad x \geq y \quad \text{use (a)} \therefore \\ x^2 + y^2 \geq xy + xy = 2xy$$

(c) The result is symmetric in x and y , so it does not matter which is larger.

3. (C SSA 2018, 13d)

$$\begin{aligned} \textcircled{a} \quad & (a+b-c)(a+c-b) \\ &= (a+b-c)(a-(b-c)) \\ &= (a^2 - (b-c)^2) \\ &\leq a^2 \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad & a^2 \geq (a+b-c)(a+c-b) \\ & b^2 \geq (b+a-c)(b+c-a) \\ & c^2 \geq (c+a-b)(c+b-a) \end{aligned}$$

$$\begin{aligned} a^2 b^2 c^2 &\geq (a+b-c)(a+c-b)(b+a-c)(b+c-a)(c+a-b)(c+b-a) \\ &= (a+b-c)^2 (b+c-a)^2 (a+c-b)^2 \end{aligned}$$

$$(abc)^2 \geq [(a+b-c)(b+c-a)(a+c-b)]^2$$

$$\therefore abc \geq (a+b-c)(b+c-a)(a+c-b).$$

4. BBHS 2018, 16b

$$\begin{aligned} \textcircled{a} \quad & (\sqrt{m} - \sqrt{n})^2 \geq 0 \therefore m - 2\sqrt{m}\sqrt{n} + n \geq 0 \\ & \therefore m + n \geq 2\sqrt{mn}. \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad & \left. \begin{aligned} m+n &\geq 2\sqrt{mn} \\ n+p &\geq 2\sqrt{np} \\ p+m &\geq 2\sqrt{pm} \end{aligned} \right\} (m+n)(n+p)(p+m) \geq 2\sqrt{mn} \cdot 2\sqrt{np} \cdot 2\sqrt{pm} \\ &= 8mnp. \end{aligned}$$

$$\begin{aligned} \textcircled{c} \quad & \frac{m}{n} + \frac{n}{p} \geq 2\sqrt{\frac{m}{n}} \sqrt{\frac{n}{p}} = 2\sqrt{\frac{m}{p}} \\ & \frac{p}{q} + \frac{q}{m} \geq 2\sqrt{\frac{p}{q}} \sqrt{\frac{q}{m}} = 2\sqrt{\frac{p}{m}} \end{aligned}$$

$$\begin{aligned} \frac{m}{n} + \frac{n}{p} + \frac{p}{q} + \frac{q}{m} &\geq 2\sqrt{\frac{m}{p}} + 2\sqrt{\frac{p}{m}} \\ &= 2\left(\sqrt{\frac{m}{p}} + \sqrt{\frac{p}{m}}\right) \\ &\geq 2 \times 2\sqrt{\sqrt{\frac{p}{m}} \times \sqrt{\frac{m}{p}}} = 4\sqrt{\frac{mp}{mp}} = 4. \end{aligned}$$

5. (Caringbah HS 2018, 15c)

$$(a+b)(a-b)^2 \geq 0$$

$$(a+b)(a^2 - 2ab + b^2) \geq 0$$

$$a^3 - 2a^2b + ab^2 + ba^2 - 2ab^2 + b^3 \geq 0$$

$$a^3 + b^3 - a^2b - ab^2 \geq 0$$

$$a^3 + b^3 \geq a^2b + ab^2$$

6. (Hornsby Girls 2018, 16b)

(a) $(\sqrt{a} - \sqrt{b})^2 \geq 0$

$$a - 2\sqrt{ab} + b \geq 0 \quad \therefore a + b \geq 2\sqrt{ab}$$

$$\therefore \frac{a+b}{2} \geq \sqrt{ab}$$

(b) $n! = 1 \times 2 \times 3 \times \dots \times n$

$$(n!)^{1/n} = (1 \times 2 \times 3 \times \dots \times n)^{1/n} \leq \frac{1+2+3+\dots+n}{n}$$

$$= \frac{n(n+1)}{2n} = \frac{n+1}{2}$$

$$\therefore n! \leq \left(\frac{n+1}{2}\right)^n$$

7. (JRAMS 2018, 12c)

(a) $(a-b)^2 \geq 0 \quad \therefore a^2 - 2ab + b^2 \geq 0 \quad \therefore a^2 + b^2 \geq 2ab$

(b) $\left(\sqrt{\frac{x}{y}} - \sqrt{\frac{y}{x}}\right)^2 \geq 0 \quad \therefore \frac{x}{y} - 2\sqrt{\frac{xy}{yx}} + \frac{y}{x} \geq 0$

$$\therefore \frac{x}{y} + \frac{y}{x} \geq 2$$

(c) $x_1 \times \frac{1}{x_1} \geq 1 = 1^2$ true for $n=1$

Assume it is true for $n=k$

$$(x_1 + x_2 + \dots + x_k) \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_k} \right) \geq k^2 \quad (*)$$

prove it for $n=k+1$

$$(x_1 + x_2 + \dots + x_k + x_{k+1}) \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_k} + \frac{1}{x_{k+1}} \right) \geq (k+1)^2 ?$$

$$(x_1 + x_2 + \dots + x_k + x_{k+1}) \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} + \frac{1}{x_{k+1}} \right)$$

$$= (x_1 + \dots + x_k) \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} \right) + x_{k+1} \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} + \frac{1}{x_{k+1}} \right) + (x_1 + \dots + x_k) \frac{1}{x_{k+1}}$$

$$= (x_1 + \dots + x_k) \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} \right) + x_{k+1} \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} \right) + (x_1 + \dots + x_k) \frac{1}{x_{k+1}} + 1$$

→ Continue (c)

$$= (x_1 + \dots + x_k) \left(\frac{1}{x_1} + \dots + \frac{1}{x_k} \right) + \left(\frac{x_1}{x_{k+1}} + \frac{x_{k+1}}{x_1} \right) + \left(\frac{x_2}{x_{k+1}} + \frac{x_{k+1}}{x_2} \right) + \dots + \left(\frac{x_k}{x_{k+1}} + \frac{x_{k+1}}{x_k} \right) + 1$$

$$\geq k^2 + \underbrace{2+2+\dots+2}_{k \text{ times}} + 1 \quad \text{using (*) and using (b)} + 1$$

$$= k^2 + 2k + 1 = (k+1)^2$$

Hence By mathematical induction it is true for all $n \geq 1$.

8. NSBHS 2018, 16b

$$a+b+c=1 \quad \text{and} \quad a+b+c \geq 3\sqrt[3]{abe}$$

$$(a) \quad \frac{1}{xy} + x + y \geq 3\sqrt[3]{\frac{1}{xy}xy} = 3$$

$$(b) \quad \frac{1}{a(a+1)} + a + a + 1 \geq 3 \quad \text{using (a)}$$

similarly

$$\frac{1}{b(b+1)} + b + b + 1 \geq 3$$

$$\frac{1}{c(c+1)} + c + c + 1 \geq 3$$

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} + 2(a+b+c) + 3 \geq 3+3+3$$

$$\frac{1}{a(a+1)} + \frac{1}{b(b+1)} + \frac{1}{c(c+1)} \geq 9-5=4 \quad \text{since } (a+b+c=1)$$

9. (Ind 2017, 12c)

$$(a) \quad a^2 + b^2 + c^2 \geq ab + bc + ca$$

$$a^2 + b^2 + d^2 \geq ab + ad + bd$$

$$a^2 + c^2 + d^2 \geq ac + ad + cd$$

$$b^2 + c^2 + d^2 \geq bc + bd + cd$$

$$3(a^2 + b^2 + c^2 + d^2) \geq 2(ab + bc + ca + ad + bd + cd)$$

$$a^2 + b^2 + c^2 + d^2 \geq \frac{2}{3}(ab + ac + ad + bc + bd + cd)$$

$$\begin{aligned}
 \textcircled{b} \quad \left(\frac{a+b+c+d}{4} \right)^2 &= \frac{(a+b+c+d)^2}{16} \\
 &= \frac{a^2+b^2+c^2+d^2 + 2(ab+ac+ad+bc+bd+cd)}{16} \\
 &\geq \frac{1}{16} \times \frac{2}{3} (ab+ac+ad+bc+bd+cd) + \frac{2}{16} (ab+ac+ad+bc+bd+cd) \\
 &= \frac{1}{6} (ab+ac+ad+bc+bd+cd).
 \end{aligned}$$

10. (JRAHS 2017, 15C).

$$\frac{1}{3}(x+y+z) \geq (xyz)^{1/3}$$

$$\textcircled{a} \quad \frac{xy+yz+zx}{3} \geq (xy \cdot yz \cdot zx)^{1/3} = (xyz)^{2/3}$$

$$\therefore xy+yz+zx \geq 3(xyz)^{2/3}$$

$$\begin{aligned}
 \textcircled{b} \quad a^3 &= 1 + (x+y+z) + (xy+yz+zx) + xyz \\
 &\geq 1 + 3(xyz)^{1/3} + 3(xyz)^{2/3} + xyz \\
 &= (1 + (xyz)^{1/3})^3
 \end{aligned}$$

$$\begin{aligned}
 a &\geq 1 + (xyz)^{1/3} \quad \therefore (xyz)^{1/3} \leq a-1 \\
 \therefore xyz &\leq (a-1)^3.
 \end{aligned}$$

11. Hornsbly GHS 2017, 16b

$$\textcircled{a} \quad \left(\frac{a}{2} - \frac{b}{2} \right)^2 \geq 0 \quad \therefore \frac{a^2}{4} - \frac{2ab}{4} + \frac{b^2}{4} \geq 0$$

$$\frac{a^2}{4} + \frac{b^2}{4} \geq \frac{2ab}{4} \quad \textcircled{1}$$

$$\frac{a^2}{4} + \frac{b^2}{4} \geq \frac{a^2}{4} + \frac{b^2}{4} \quad \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} \quad \therefore \frac{a^2}{2} + \frac{b^2}{2} \geq \left(\frac{a+b}{2} \right)^2$$

$$a+b \geq 2\sqrt{ab} \quad \therefore \frac{a+b}{2} \geq \sqrt{ab} \quad \therefore \left(\frac{a+b}{2} \right)^2 \geq ab.$$

$$\therefore ab \leq \left(\frac{a+b}{2} \right)^2 \leq \frac{a^2+b^2}{2}.$$

(b)

$$a+b=1$$

$$ab \leq \left(\frac{1}{2}\right)^2 = \frac{1}{4} \quad \therefore \frac{1}{ab} \geq 4$$

$$\begin{aligned} \left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 &\geq 2 \left(\frac{a + \frac{1}{a} + b + \frac{1}{b}}{2} \right)^2 \\ &= 2 \left(\frac{a+b + \frac{a+b}{ab}}{2} \right)^2 \\ &= 2 \left(\frac{1+4}{2} \right)^2 = \frac{25}{2} \end{aligned}$$

Q. Ascham 2016, 12b

$$(a) \quad \frac{a}{b} + \frac{b}{a} \geq 2 \sqrt{\frac{a}{b} \frac{b}{a}} = 2$$

$$(b) \quad \left. \begin{aligned} p^2 + q^2 &\geq 2pq \\ p^2 + r^2 &\geq 2pr \\ q^2 + r^2 &\geq 2qr \end{aligned} \right\} p^2 + q^2 + r^2 \geq pq + qr + rp$$

$$\begin{aligned} (c) \quad p^3 + q^3 &= (p+q)(p^2 + q^2 - pq) \\ &\geq (p+q)(2pq - pq) \\ &= pq(p+q) \end{aligned}$$

$$\begin{aligned} (d) \quad p^3 + q^3 &\geq pq(p+q) \\ p^3 + r^3 &\geq pr(p+r) \\ q^3 + r^3 &\geq qr(q+r) \end{aligned}$$

$$2(p^3 + q^3 + r^3) \geq pq(p+q) + pr(p+r) + qr(q+r)$$

13. (Gringbani 2016, 16c)

$$a+b \geq 2\sqrt{ab}$$

$$a+b+c=1$$

$$\frac{1}{a} - 1 = \frac{1-a}{a} = \frac{b+c}{a}, \quad \frac{1}{b} - 1 = \frac{a+c}{b}, \quad \frac{1}{c} - 1 = \frac{a+b}{c}$$

$$\begin{aligned} \left(\frac{1}{a} - 1\right) \left(\frac{1}{b} - 1\right) \left(\frac{1}{c} - 1\right) &= \frac{b+c}{a} \times \frac{a+c}{b} \times \frac{a+b}{c} \geq \frac{2\sqrt{bc}}{a} \times \frac{2\sqrt{ac}}{b} \times \frac{2\sqrt{ab}}{c} \\ &= 8 \end{aligned}$$

14. Hunters Hill 2016, 14b

$$(a) \quad (a-b)^2 \geq 0 \therefore a^2 - 2ab + b^2 \geq 0 \therefore a^2 + b^2 \geq 2ab$$

$$(b) \quad p+2 \geq 2\sqrt{2p}$$

$$q+2 \geq 2\sqrt{2q}$$

$$p+q \geq 2\sqrt{pq}$$

$$(p+2)(q+2)(p+q) \geq 8\sqrt{2p2qpq} = 16pq$$

15. Ind 2016, 13b

$$(a) \quad (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$a^2 + b^2 \geq 2ab$$

$$a^2 + c^2 \geq 2ac$$

$$b^2 + c^2 \geq 2bc$$

$$\left. \begin{array}{l} a^2 + b^2 \geq 2ab \\ a^2 + c^2 \geq 2ac \\ b^2 + c^2 \geq 2bc \end{array} \right\} a^2 + b^2 + c^2 \geq ab + bc + ca$$

$$\therefore (a+b+c)^2 \geq 3(ab+bc+ca)$$

$$(b) \quad \text{use (a)}$$

$$(ab+bc+ca)^2 \geq 3(ab^2c + bc^2a + a^2bc) \\ = 3abc(a+b+c)$$

$$(c) \quad (a+b+c)^3 = (a+b+c)(a+b+c)^2 \\ \geq 3(a+b+c)(ab+bc+ca)$$

$$(a) \text{ \& (b) } \therefore [(a+b+c)(ab+bc+ca)] \geq 9_{abc} (a+b+c)(ab+bc+ca)$$

$$(a+b+c)(ab+bc+ca) \geq 9abc$$

$$(a+b+c)^3 \geq 3 \times 9abc = 27abc$$

16. (Knox Grammar 2016, 13d)

$$(a) \quad (p+q)^2 = (p-q)^2 + 4pq \geq 4pq$$

$$p+q \geq 2\sqrt{pq}$$

$$\frac{p+q}{2} \geq \sqrt{pq}$$

$$(b) \quad \left. \begin{array}{l} \frac{p+q}{2} \geq \sqrt{pq} \\ \frac{r+s}{2} \geq \sqrt{rs} \end{array} \right\} \frac{\frac{p+q}{2} + \frac{r+s}{2}}{2} \geq \sqrt{\frac{p+q}{2} \times \frac{r+s}{2}} \\ \geq \sqrt{\sqrt{pq} \times \sqrt{rs}} \\ = \sqrt[4]{pqrs}$$

$$\therefore \frac{p+q+r+s}{4} \geq \sqrt[4]{pqrs}$$

17. (NSBHS 2016, 16a)

$$(a) \quad \frac{a+b}{2} \geq \sqrt{ab}$$

$$\frac{c+d}{2} \geq \sqrt{cd}$$

$$\frac{\frac{a+b}{2} + \frac{c+d}{2}}{2} \geq \frac{\sqrt{ab} + \sqrt{cd}}{2} \geq \sqrt{\sqrt{ab}\sqrt{cd}} = \sqrt[4]{abcd}$$

$$(b) \quad \frac{a+b+c}{3} = \frac{1}{4} \left(a+b+c + \frac{a+b+c}{3} \right) \geq \sqrt[4]{abc \left(\frac{a+b+c}{3} \right)}$$

$$\therefore \left(\frac{a+b+c}{3} \right)^4 \geq abc \left(\frac{a+b+c}{3} \right)$$

$$\therefore \left(\frac{a+b+c}{3} \right)^3 \geq abc$$

$$\therefore \frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

18. (NSGHS 2016, 14C)

(a) $(a-b)^2 \geq 0 \therefore a^2 - 2ab + b^2 \geq 0 \therefore a^2 + b^2 \geq 2ab$

(b) $a^2 + b^2 \geq 2ab$

$$a^2 + c^2 \geq 2ac$$

$$b^2 + c^2 \geq 2bc$$

$$\therefore a^2 + b^2 + a^2 + c^2 + b^2 + c^2 \geq 2ab + 2ac + 2bc$$

$$\therefore 2(a^2 + b^2 + c^2) \geq 2(ab + ac + bc)$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca$$

(c) $x^3 - 3x^2 + 4x + k = 0$

$$\alpha + \beta + \gamma = 3$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 4$$

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= 3^2 - 2(4) = 1\end{aligned}$$

But $\alpha^2 + \beta^2 + \gamma^2 \geq \alpha\beta + \beta\gamma + \gamma\alpha$ if α, β and γ are real
(see part b)

$$\therefore 1 \geq 4 \text{ Contradiction}$$

$\therefore \alpha, \beta$ and γ Can not be all real.

19. (Penrith HS 2016, 14d)

If $0 < x < y < \frac{1}{2}$

$$(\sqrt{x} - \sqrt{y})^2 \geq 0 \therefore x + y - 2\sqrt{xy} \geq 0 \therefore \frac{x+y}{2} \geq \sqrt{xy}$$

$$\therefore x + y > \frac{x+y}{2} \geq \sqrt{xy}$$

$$x \leq \frac{1}{2} \text{ and } y \leq \frac{1}{2} \therefore x + y \leq 1 \therefore \sqrt{x+y} \geq \sqrt{xy}$$

as the square root of a number between 0 and 1 is greater than the number.

$$\therefore \sqrt{xy} < x + y < \sqrt{x+y}$$

20. (Sam al Hosni 2016, 15a)

$$\begin{aligned} \textcircled{a} \quad & (a-b)^2 + (a-c)^2 + (b-c)^2 \geq 0 \\ & a^2 - 2ab + b^2 + a^2 - 2ac + c^2 + b^2 - 2bc + c^2 \geq 0 \\ & 2(a^2 + b^2 + c^2) - 2(ab + bc + ca) \geq 0 \\ & \therefore a^2 + b^2 + c^2 \geq ab + bc + ca \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad & \frac{p^2}{q^2} + \frac{q^2}{r^2} + \frac{r^2}{p^2} \geq \frac{p}{q} \frac{q}{r} + \frac{p}{r} \frac{r}{p} + \frac{q}{r} \frac{r}{p} \\ & = \frac{p}{r} + \frac{r}{q} + \frac{q}{p} \end{aligned}$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

a, b, c are positive $\therefore a+b+c \geq 0$
and from $\textcircled{a}$ $a^2 + b^2 + c^2 - ab - bc - ac \geq 0$.

$$\therefore a^3 + b^3 + c^3 - 3abc \geq 0$$

$$\text{let } a = \sqrt[3]{\frac{p}{r}}, b = \sqrt[3]{\frac{q}{p}}, c = \sqrt[3]{\frac{r}{q}}$$

$$\therefore \frac{p}{r} + \frac{q}{p} + \frac{r}{q} - 3\sqrt[3]{\frac{p}{r} \frac{q}{p} \frac{r}{q}} \geq 0$$

$$\therefore \frac{p}{r} + \frac{q}{p} + \frac{r}{q} \geq 3$$

21. (western region 2016, 14d)

$$\textcircled{a} \quad (x-y)^2 \geq 0 \therefore x^2 - 2xy + y^2 \geq 0 \therefore x^2 + y^2 \geq 2xy$$

$$\left. \begin{array}{l} x^2 + y^2 \geq 2xy \\ y^2 + z^2 \geq 2yz \\ x^2 + z^2 \geq 2xz \end{array} \right\} \begin{array}{l} x^2 + y^2 + y^2 + z^2 + z^2 + x^2 \geq 2xy + 2xy + 2yz \\ \therefore x^2 + y^2 + z^2 \geq xy + yz + zx \end{array}$$

$$\textcircled{b} \quad x+y+z=3$$

$$(x+y+z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx) \geq 3(xy + yz + zx)$$

$$9 \geq 3(xy + yz + zx) \therefore xy + yz + zx \leq 3$$

22. (ACE 2015, 16b)

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$\text{but } ab+bc+ca \leq a^2+b^2+c^2$$

$$\therefore (a+b+c)^2 \leq 3(a^2+b^2+c^2)$$

23. (Ind 2015, 16c)

(a) $a > b > c > 1$

$$a^{a-b} b^{b-c} > c^{a-b} c^{b-c} = c^{a-c} \quad \text{since } a > b > c > 1$$

$$(a^b b^c c^c) a^{a-b} b^{b-c} > a^b b^c c^c c^{a-c}$$

$$a^a b^b c^c > a^b b^c c^a$$

(b) since $f(x) = \ln(x)$ is monotonic increasing function,
 $\ln(a^a b^b c^c) > \ln(a^b b^c c^a)$

$$\ln a^a + \ln b^b + \ln c^c > \ln a^b + \ln b^c + \ln c^a$$

$$\therefore a \ln a + b \ln b + c \ln c > b \ln a + c \ln b + a \ln c$$

24. (Knox Grammar 2015, 15c)

(a) $a^2 + b^2 \geq 2ab, a^2 + c^2 \geq 2ac, a^2 + d^2 \geq 2ad$

$$b^2 + c^2 \geq 2bc, b^2 + d^2 \geq 2bd \quad \text{and } c^2 + d^2 \geq 2cd$$

$$3(a^2 + b^2 + c^2 + d^2) \geq 2(ab+ac+bc+ad+bd+cd)$$

(b) $(a+b+c+d)^2 = a^2 + b^2 + c^2 + d^2 + 2(ab+ac+bc+ad+bd+cd)$

$$3(a+b+c+d)^2 = 3(a^2 + b^2 + c^2 + d^2) + 6(ab+ac+bc+ad+bd+cd)$$

$$\geq 2(ab+ac+bc+ad+bd+cd) + 6(ab+ac+bc+ad+bd+cd)$$

$$= 8(ab+ac+bc+ad+bd+cd)$$

$$\therefore ab+ac+ad+bc+bd+cd \leq \frac{3}{8} (a+b+c+d)^2 = \frac{3}{8}$$

$$\text{since } a+b+c+d = 1.$$

25. (NSGHS 2015, 16b)

(a) $(a-b)^2 \geq 0 \therefore a^2 - 2ab + b^2 \geq 0 \therefore a^2 + b^2 \geq 2ab$

(b) $a^4 + b^4 \geq 2a^2b^2$
 $c^4 + d^4 \geq 2c^2d^2$

$$a^4 + b^4 + c^4 + d^4 \geq 2(a^2b^2 + c^2d^2) \\ \geq 2 \times 2 \sqrt{a^2b^2c^2d^2} = 4abcd$$

26. (Sam El Hosri 2015, 16c)

$$a+b+c \geq 3\sqrt[3]{abc}$$

$$(p+2)(q+2)(r+2) = (p+2)(qr+2q+2r+4) \\ = pqr + 2pq + 2pr + 2qr + 4(p+q+r) + 8$$

$$64 = pqr + 2(pq + pr + qr) + 4(p+q+r) + 8 \\ \geq pqr + 6\sqrt[3]{p^2q^2r^2} + 12\sqrt[3]{pqr} + 8$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

let $a = \sqrt[3]{pqr}$ and $b = 2$

$$(\sqrt[3]{pqr} + 2)^3 = pqr + 6\sqrt[3]{p^2q^2r^2} + 12\sqrt[3]{pqr} + 8$$

$$\therefore 64 \geq (\sqrt[3]{pqr} + 2)^3$$

$$\therefore 4 \geq \sqrt[3]{pqr} + 2$$

$$\therefore 2 \geq \sqrt[3]{pqr}$$

$$\therefore pqr \leq 8$$

27. (Ind 2014, 15b)

(a) $(\sqrt{a} - \frac{1}{\sqrt{a}})^2 \geq 0 \therefore a - 2\sqrt{a} \times \frac{1}{\sqrt{a}} + \frac{1}{a} \geq 0$

$$\therefore a + \frac{1}{a} \geq 2$$

(b)(i) $\frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c}$

$$= \frac{b}{a} + \frac{c}{a} + \frac{c}{b} + \frac{a}{b} + \frac{a}{c} + \frac{b}{c}$$

$$= \left(\frac{b}{a} + \frac{a}{b}\right) + \left(\frac{c}{a} + \frac{a}{c}\right) + \left(\frac{c}{b} + \frac{b}{c}\right)$$

$$\geq 2 + 2 + 2 \quad (\text{using part a})$$

$$= 6$$

(ii) use part (i)

Replacing $a \rightarrow b+c$

$b \rightarrow c+a$

and $c \rightarrow a+b$

$$\frac{(c+a) + (a+b)}{b+c} + \frac{(a+b) + (b+c)}{c+a} + \frac{(b+c) + (c+a)}{a+b} \geq 6$$

$$1 + \frac{2a}{b+c} + 1 + \frac{2b}{c+a} + 1 + \frac{2c}{a+b} \geq 6$$

$$2 \left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \right) \geq 3$$

$$\therefore \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

28. (SBHS 2014, 14C)

(a) $(x+y)^2 = x^2 + y^2 + 2xy \geq 2xy + 2xy = 4xy$
since $(x-y)^2 \geq 0 \Rightarrow x^2 + y^2 \geq 2xy$.

(b) $[(x+y)(z+w)]^2 = (x+y)^2 (z+w)^2$
 $\geq 4xy \times 4zw = 16xyzw$

(c) $\frac{x+y+z+w}{4} = \frac{\frac{x+y}{2} + \frac{z+w}{2}}{2} \geq \frac{\sqrt{xy} + \sqrt{zw}}{2}$ using (a)
 $\geq \sqrt{\sqrt{xy} \sqrt{zw}}$
 $= \sqrt[4]{xyzw}$

(d) $\frac{1}{4} \left(\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \right) \geq \sqrt[4]{\frac{a}{b} \frac{b}{c} \frac{c}{d} \frac{d}{a}} = 1$ using (c)

$\therefore \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4$.

29. (ACE 2013, 16C)

$a^2 + b^2 = (a-b)^2 + 2ab \geq (a-b)^2 + 2$ since $ab > 1$

$\frac{a^2 + b^2}{a-b} \geq a-b + \frac{2}{a-b}$

$\geq 2\sqrt{(a-b) \frac{2}{a-b}} = 2\sqrt{2}$.

30. (a) $a^2 - ab + b^2 = \frac{1}{4}(a+b)^2 + \frac{3}{4}(a-b)^2$ where
 $\frac{3}{4}(a-b)^2 \geq 0$ since a and b are real.

$$a^2 - ab + b^2 \geq \left(\frac{a+b}{2}\right)^2 \text{ with equality iff } a=b.$$

(b) using cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos \angle BCA$$

If $\angle BCA \geq 60^\circ \therefore \cos \angle BCA \leq \frac{1}{2}$

$$\therefore c^2 \geq a^2 + b^2 - 2ab \times \frac{1}{2}$$

$$c^2 \geq a^2 + b^2 - ab$$

with equality iff $\angle BCA = 60^\circ$.

using part (a)

$$c \geq \sqrt{a^2 - ab + b^2} \geq \frac{a+b}{2} \text{ with equality if}$$

$\angle BCA = 60^\circ$ and $a=b$ (equilateral triangle).

$$\therefore c^3 \geq \left(\frac{a+b}{2}\right)(a^2 + b^2 - ab) = \frac{a^3 + b^3}{2}$$

$\therefore 2c^3 \geq a^3 + b^3$ with equality iff $\triangle ABC$ is an equilateral triangle.

31 (NHBHS 2013, 166)

(a) $(\sqrt{k+2} - \sqrt{k+1})^2 \geq 0$

$$k+2 + k+1 - 2\sqrt{(k+2)(k+1)} \geq 0$$

$$2k+3 \geq 2\sqrt{k+2}\sqrt{k+1}$$

(b) $1 \geq 2\sqrt{k+2}\sqrt{k+1} - 2(k+1)$ (Use (a))

$$\frac{1}{\sqrt{k+1}} \geq 2\sqrt{k+2} - 2\sqrt{k+1}$$

$$\frac{1}{\sqrt{k+1}} \geq 2(\sqrt{k+2} - \sqrt{k+1})$$

(c) $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} = \sum_{k=0}^{n-1} \frac{1}{\sqrt{k+1}}$

$$\geq 2 \sum_{k=0}^{n-1} (\sqrt{k+2} - \sqrt{k+1})$$

$$= 2(\sqrt{2} - \sqrt{1} + \sqrt{3} - \sqrt{2} + \sqrt{4} - \sqrt{3} + \dots + \sqrt{n+1} - \sqrt{n})$$
$$= 2(\sqrt{n+1} - 1)$$

32 (NSGHS 2013, 13a)

(a) $x^2 + y^2 \geq 2xy > xy$

(b) $x+y = 3z$

$$(x+y)^2 = x^2 + y^2 + 2xy \leq x^2 + y^2 + 2(x^2 + y^2)$$
$$= 3(x^2 + y^2)$$

$$(3z)^2 \leq 3(x^2 + y^2)$$

$$3z^2 \leq x^2 + y^2$$

33 (Pennith 2013, 15d)

(a) $x^2 + x + 1 \geq 0$

$\Delta = 1 - 4 = -3 < 0$. $a > 0$. concave up.

$\therefore x^2 + x + 1 > 0$ for all real x .

(b) let $x = \frac{a}{b}$

$\left(\frac{a}{b}\right)^2 + \frac{a}{b} + 1 \geq 0$

$a^2 + ab + b^2 \geq 0$

34 (Sam d Hosni 2013, 15a)

(a) $\left(\sqrt{a} + \frac{1}{\sqrt{a}}\right)^2 \geq 0 \Rightarrow a + \frac{1}{a} \geq 2$

(b) $a + b + c = 1$

$$\left. \begin{array}{l} a + \frac{1}{a} \geq 2 \\ b + \frac{1}{b} \geq 2 \\ c + \frac{1}{c} \geq 2 \end{array} \right\} \begin{array}{l} a + b + c + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 6 \\ \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 5 \text{ since } a+b+c=1 \end{array}$$

35 (NSGHS 2012, 15c)

(a) $(a-b)^2 \geq 0 \therefore a^2 + b^2 \geq 2ab$

(b) $\tan^2 \theta + \sec^2 \theta + \csc^2 \theta$

$\tan^2 \theta + \sec^2 \theta \geq 2 \tan \theta \sec \theta = 2 \sin \theta$

$\tan^2 \theta + \csc^2 \theta \geq 2 \tan \theta \csc \theta = 2 \sec \theta$

$\sec^2 \theta + \csc^2 \theta \geq 2 \sec \theta \csc \theta = 2 \cot \theta$

$\therefore \tan^2 \theta + \sec^2 \theta + \csc^2 \theta \geq \sin \theta + \sec \theta + \cot \theta$

36. (SBHS 2012, 15b)

(a) $(\sqrt{x} - \sqrt{y})^2 \geq 0 \therefore x - 2\sqrt{xy} + y \geq 0 \therefore \frac{x+y}{2} \geq \sqrt{xy}$

(b) let $x = \frac{1}{a}$ and $y = \frac{1}{b}$

$$\frac{\frac{1}{a} + \frac{1}{b}}{2} \geq \sqrt{\frac{1}{ab}} \therefore \frac{a+b}{2ab} \geq \frac{1}{\sqrt{ab}}$$

$$\therefore \frac{2ab}{a+b} \leq \sqrt{ab}$$

(c) $\frac{1}{n-1} + \frac{1}{n} + \frac{1}{n+1} \geq \frac{1}{n} + \frac{2}{\sqrt{x^2-1}}$

using (b) $\frac{2(x^2-1)}{2n} \leq \sqrt{x^2-1} \therefore \frac{2\sqrt{x^2-1}}{2n} \leq 1 \therefore \frac{1}{\sqrt{x^2-1}} \geq \frac{1}{x}$

$$\therefore \frac{1}{n-1} + \frac{1}{n} + \frac{1}{n+1} \geq \frac{1}{n} + \frac{2}{x} = \frac{3}{x}$$

(d) $H = 1 + \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7}\right) + \dots + \left(\frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n}\right)$

$$\geq 1 + \frac{3}{3} + \frac{3}{6} + \frac{3}{9} + \dots + \frac{3}{n-1} \quad \text{using (c)}$$

$$= 1 + 1 + \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) + \dots + \frac{1}{k} \quad (\text{letting } 3k = n-1)$$

$$\geq 1 + 1 + 1 + \dots + \frac{3}{k-1} \quad (\text{let } k-1 = 3m)$$

$$\geq 1 + 1 + 1 + \dots + \frac{1}{m}$$

$$H \geq 1 + 1 + 1 + \dots + \frac{1}{m}$$

As $n \rightarrow \infty$, $H \geq 1 + 1 + 1 + \dots$

$\therefore H$ has no limit.

37 (Manly SC 2011, 6b)

(a) $a^2 + b^2 \geq 2ab$

(b) $a^2 + b^2 + c^2 \geq ab + bc + ca$

(c) $\sin^2 \alpha + \cos^2 \alpha \geq 2 \sin \alpha \cos \alpha = \sin 2\alpha$

(d) $\sin^2 \alpha + \cos^2 \alpha + \tan^2 \alpha$

$$\sin^2 \alpha + \cos^2 \alpha \geq \sin 2\alpha$$

$$\cos^2 \alpha + \tan^2 \alpha \geq 2 \sin \alpha$$

$$\sin^2 \alpha + \tan^2 \alpha \geq 2 \sin \alpha \sec \alpha = 2(1 - \cos^2 \alpha) \sec \alpha = 2 \sec \alpha - 2 \cos \alpha$$

$$\sin^2 \alpha + \cos^2 \alpha + \cos^2 \alpha + \tan^2 \alpha + \tan^2 \alpha + \sin^2 \alpha \geq \sin 2\alpha + 2 \sec \alpha + 2 \sec \alpha - 2 \cos \alpha$$

$$\sin^2 \alpha + \cos^2 \alpha + \tan^2 \alpha \geq \frac{1}{2} (\sin 2\alpha + 2 \sec \alpha + 2 \sec \alpha - 2 \cos \alpha)$$

$$= \sin \alpha - \cos \alpha + \sec \alpha + \frac{1}{2} \sin 2\alpha.$$

38 (NSEHS 2011, 8b)

(a) $a^2 + b^2 \geq 2ab$, $a^2 + b^2 + c^2 \geq ab + bc + ca$

(b) $a + b + c = 9$

$$ab + bc + ca \leq 27$$

$$(abc) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) = bc + ac + ab \leq 27$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{27}{abc}$$

39 (C SSA 2009, 6b).

(a) $x + y \geq 2\sqrt{xy}$

(b) $x + y + z + w \geq 4\sqrt[4]{xyzw}$

(c) $x + y + z + \frac{x + y + z}{3} \geq 4\sqrt[3]{xyz \left(\frac{x + y + z}{3} \right)}$

$$\therefore \left(\frac{x + y + z}{3} \right)^4 \geq xyz \left(\frac{x + y + z}{3} \right)$$

$$\therefore \left(\frac{x + y + z}{3} \right)^3 \geq xyz$$

$$\therefore \frac{x + y + z}{3} \geq \sqrt[3]{xyz}$$